

A Structure-Revealing Tensor Network Paradigm and Its Applications

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Abstract—Tensor network (TN) decomposition is a powerful data representation tool widely used in machine learning and computer vision. However, existing TN decomposition methods are limited by fixed topologies, which constrain their representational capacity. To overcome this limitation, we introduce, for the first time, an energy-based framework that allows us to reveal a data-adaptive TN structure through the optimization of an energy function. Specifically, we propose a novel structure-revealing TN paradigm, termed SVD-inspired TN decomposition (SVDinsTN). This paradigm enables the formulation of an energy function that simultaneously considers representation error and TN structure complexity, facilitating the joint optimization of data representation and the identification of a compact TN structure. Theoretically, we present the principles behind the energy function and derive an upper bound for the TN rank. In contrast to heuristic TN structure search methods, our energy-based method achieves a remarkable acceleration of 100 to 1000 times, making it practical for real-world applications. Experimental results in neural network compression, remote sensing image cloud removal, and tensor robust principal component analysis consistently show that SVDinsTN can identify a TN structure that better aligns with the given data, achieving superior performance over fixed-topology methods.

Index Terms—Tensor network decomposition, tensor network structure search, low-rank representation, energy function

1 INTRODUCTION

A higher-order tensor is a multidimensional array, which is a mathematical tool to represent higher-order data in the real world, such as color videos, remote sensing images (RSIs), light field images (LFIs), and traffic data. How to capture the intrinsic information of higher-order data and represent such data with few parameters is a central issue in various areas of signal and image processing, machine learning, computer vision, etc. Tensor decomposition, which represents a higher-order tensor by a series of low-dimensional factors with a specific operation among them, is a powerful tool for characterizing the global correlations of higher-order data [1], [2], [3], [4], [5], [6]. In the past decade, CANDECOMP/PARAFAC (CP) decomposition [7], [8], Tucker decomposition [9], [10], tensor singular value decomposition (t-SVD) [11], [12], [13], and their variants [10], [14], [15], [16], [17], [18] have been widely applied in many

applications, such as RSI restoration, LFI compression, and traffic flow prediction, and have achieved promising results [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29].

Inspired by concepts from physics [30], [31], [32], tensor network (TN) decompositions have recently attracted increasing attention for representing higher-order tensors [33], [34], [35]. A TN decomposition represents an N th-order tensor by a collection of low-order factor matrices or tensors, referred to as *TN cores*. These cores are connected through tensor contractions, and their interactions jointly determine the represented tensor. By treating TN cores as nodes and contractions between them as edges, a TN can be described by an unweighted graph, referred to as the *TN topology*. The dimensions associated with the edges are collectively called the *TN rank*. Together, the TN topology and TN rank determine the *TN structure*. Different choices of TN topology give rise to different TN decompositions. For example, tensor train (TT) [36], tensor ring (TR) [37], and fully-connected tensor network (FCTN) [38] correspond to chain, ring, and fully-connected graph structures, respectively. Fig. 1 shows their graphical representations for a fifth-order tensor.

TT decomposition represents an N th-order tensor by a chain of TN cores, consisting of two boundary matrices and $N-2$ third-order tensors, with contractions performed only between neighboring cores. Within the framework of TT decomposition, Bengua et al. proposed two high-dimensional image completion models with block coordinate descent (BCD)-based solvers [39], Yang et al. introduced a tensor robust principal component analysis (TRPCA) model with an alternating direction method of multipliers (ADMM)-based solver [40], and Dian et al. developed a hyperspectral image (HSI) super-resolution model with an ADMM-based solver [41]. TR decomposition represents an N th-order tensor by a cyclic sequence of N third-order TN cores, where

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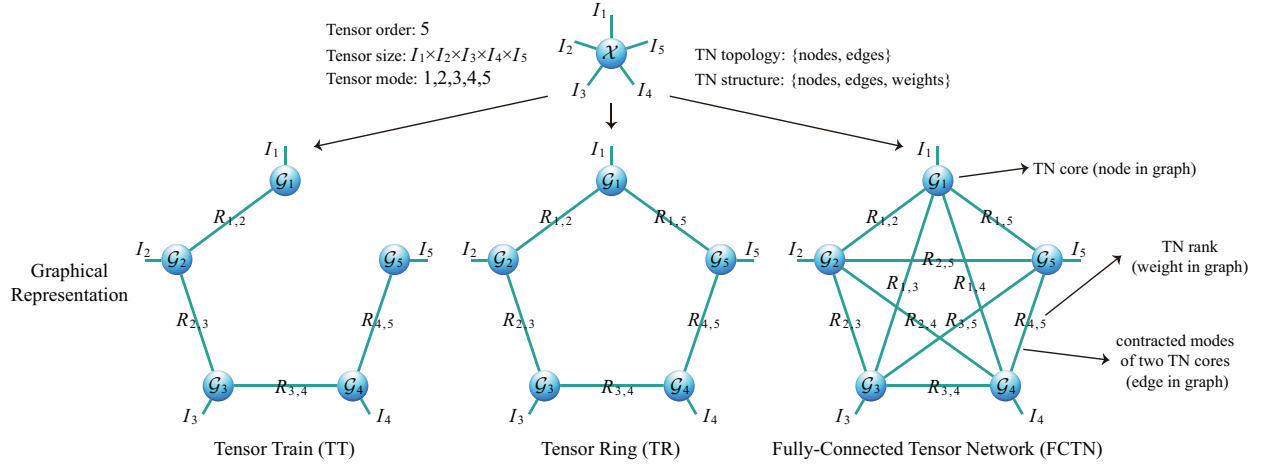


Fig. 1. A graphical representation of TT, TR, and FCTN decompositions on a fifth-order tensor.

TABLE 1
Representation performance of different TN structures with all ranks fixed at 2. The reported representation error is averaged over 500 independent tests.

Structure of test tensors	Structure of TN decompositions		
Representation error	0.06064	0.00083	0.00295

the first and last cores are connected. Within the framework of TR decomposition, Qiu et al. introduced a noisy tensor completion model and employed an ADMM-based algorithm [42]. He et al. proposed a remote sensing image reconstruction model and developed an efficient algorithm by combining the alternating least squares (ALS) method with the augmented Lagrange multiplier method [43]. Wang et al. proposed a deep neural network compression method [44]. However, the “chain” and “ring” topologies impose inherent limitations on TT and TR decompositions. First, since each core is directly connected only to its neighboring cores, these models may be insufficient to characterize long-range dependencies among different tensor modes. Second, the representation ability of TT- and TR-based methods is sensitive to mode permutations, as different mode orders may produce different representations [45].

To address the aforementioned limitations of TT and TR decompositions, Zheng et al. [38] proposed the FCTN decomposition, which represents an N th-order tensor by N small-scale N th-order TN cores with pairwise contractions among all cores. Within the framework of FCTN decomposition, Zheng et al. proposed a multispectral image inpainting model and designed a proximal alternating minimization (PAM)-based algorithm [46], Liu et al. introduced a TRPCA model with an ADMM-based algorithm and established its exact recovery guarantee [47], and Li et al. proposed a convolutional neural network compression method for HSI classification [48]. Different from TT and TR decompositions, FCTN decomposition allows pairwise

contractions among all TN cores. Such a “fully-connected” topology strengthens its ability to model global correlations across tensor modes and makes the resulting representation invariant to mode permutations.

The topologies of the above-mentioned TN decomposition methods are predefined and fixed. In the real world, however, each tensor is unique with specific structure and characteristics. If a tensor has a simple structure, using a TN decomposition with a complex topology may introduce unnecessary computational complexity and lead to overfitting, resulting in an inefficient representation. In contrast, if a tensor has a complex structure, using a TN decomposition with a simple topology may not be adequate to capture the complex structural information of the tensor, leading to degraded performance (see Tab. 1). Therefore, the choice of topology for a TN decomposition should be tailored to the specific structure and characteristics of the tensor, and selecting the most suitable topology can result in a more accurate and efficient TN decomposition. In addition to TN topology, determining an appropriate TN rank is also difficult in practical applications. Recently, researchers have imposed various regularization techniques, such as the sum of nuclear norms, Tikhonov regularization, and group sparsity constraints, on TN cores to reduce the impact of the selection of TN rank [49], [50], [51]. However, this strategy mainly alleviates rank-selection sensitivity in particular tensor recovery tasks, rather than fundamentally addressing the joint determination of TN topology and rank.

TN structure search (TN-SS) aims to identify an appropriate TN structure, including both TN topology and TN rank, to achieve a compact and efficient representation of a given tensor, thereby enhancing storage and computational efficiency [45], [52], [53], [54], [55], [56]. For the challenging TN-SS problem, the sampling-evaluation framework provides a straightforward method, which involves generating various candidate TN structures (sampling) and subsequently evaluating their effectiveness or performance in representing the given data or accomplishing specific tasks (evaluation). Different sampling strategies play a critical role in determining the search efficiency and performance of TN-SS methods. The choice of sampling approach influences both the exploration of candidate structures and the cover-

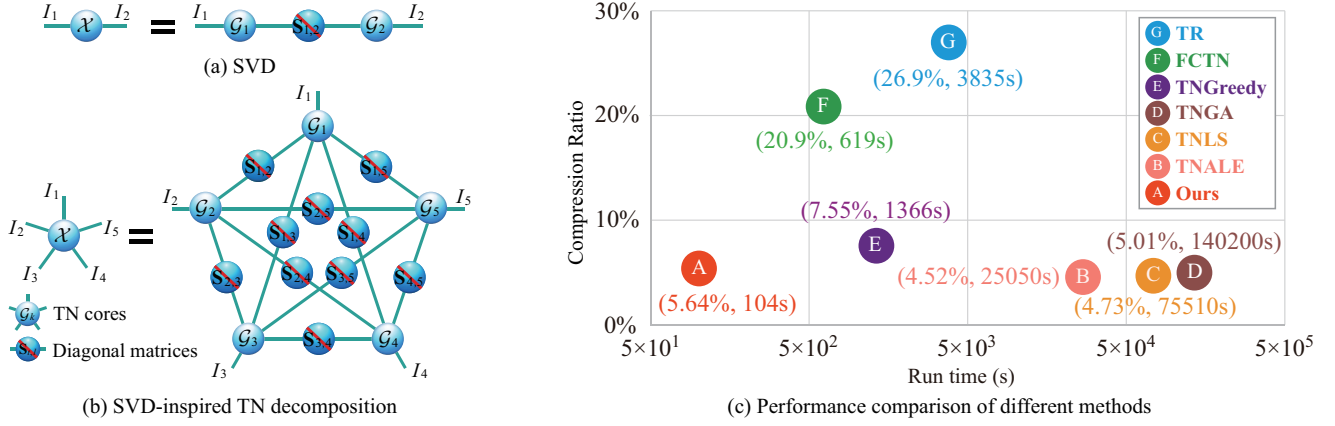


Fig. 2. (a) A graphical illustration of SVD. (b) A graphical illustration of SVD-inspired TN decomposition on a fifth-order tensor. (c) Comparison of compression ratio ($\downarrow$) and run time ($\downarrow$) for different methods on the light field image *Knights* under an error bound of 0.05, where TR [37] and FCTN [38] are fixed-topology TN decomposition methods, and TNGreedy [52], TNGA [53], TNLS [45], and TNALE [54] are TN-SS methods.

age of the search space, thereby striking a balance between computational cost and the quality of the identified TN structures. For instance, Hashemizadeh et al. [52] adopted an arithmetic increasing sampling method, initializing with the simplest TN structure. While this approach offers higher search efficiency, it tends to be unstable and susceptible to being trapped in local minima. In contrast, Li et al. [53] employed a genetic-algorithm-inspired sampling method, which starts with a set of random TN structures. This method is effective at avoiding local minima, but it requires sampling a large number of TN structures, making it significantly more time-consuming. However, regardless of how advanced the sampling strategy is, the sampling-evaluation framework inevitably requires numerous repeated structure evaluations, which makes it computationally expensive and limits its practical applications. Therefore, *a non-trivial challenge is to directly construct a structure-revealing TN decomposition paradigm to bypass the heuristic sampling-evaluation framework, thereby reducing the computational burden.*

In this paper, we construct a structure-revealing TN paradigm for the first time, termed SVD-inspired TN decomposition (SVDinsTN). SVDinsTN inserts a series of diagonal matrices between any pair of TN cores in FCTN decomposition and reveals a compact TN structure by promoting the sparsity of the inserted diagonal matrices (see Fig. 2(b)). The intuition behind SVDinsTN is inspired by the matrix SVD, where the non-zero diagonal entries determine the matrix rank. Similarly, the non-zero entries of the inserted diagonal matrices in SVDinsTN determine the FCTN rank. Since an FCTN rank corresponds to a unique TN structure (see Property 1), sparsifying these diagonal matrices enables SVDinsTN to reveal a compact TN structure.

Based on SVDinsTN, we construct an energy function that jointly considers the representation error and the complexity of the TN structure. By minimizing this energy function, the TN cores and diagonal matrices are iteratively updated, and a sparsity-inducing function is imposed on the diagonal matrices to reveal a compact TN structure. This directly avoids repeated sampling and evaluation of candidate TN structures. Notably, the proposed method does not require specially designed sparsity-inducing functions or dedicated solvers. Instead, the standard ℓ_1 -norm or ℓ_0 -

norm, combined with proximal alternating minimization, is sufficient to achieve significantly higher computational efficiency than state-of-the-art (SOTA) TN-SS methods, with up to 1000-fold acceleration (see Fig. 2(c)).

To demonstrate the practical applicability of SVDinsTN in real-world scenarios, we apply it to neural network compression, remote sensing image cloud removal, and tensor robust principal component analysis. Experimental results show that SVDinsTN can identify a data-adaptive TN structure and achieve superior performance over existing fixed-topology methods.

In the proposed framework, a major challenge is the high computational cost in the initial iterations, which arises from using a “fully-connected” topology as the initial configuration. To address it, we first theoretically establish an upper bound for the TN rank and then design an initialization scheme. This scheme restricts the rank on each edge of the initial TN topology to a smaller range, thereby effectively reducing the computational cost in the initial iterations.

This article extends the material published at CVPR [57]. The new contributions are as follows: (1) We establish the theoretical foundations of the SVDinsTN paradigm, encompassing its modeling and computational framework, offering profound insights to enhance understanding and optimize practical implementations (see Sections 3.1 and 3.2). (2) Moving beyond the ℓ_1 -norm, we construct a generalized energy function that facilitates the exploration of more effective sparsity-inducing functions, driving further performance improvements. Furthermore, we analyze the influence of various sparsity-inducing functions in real-world applications (see Sections 3.2 and 5). (3) We present foundational SVDinsTN-based models tailored for diverse real-world tasks, such as neural network compression, remote sensing image cloud removal, and tensor robust principal component analysis, highlighting its versatility and effectiveness as a core tool for these applications (see Section 4).

The outline of this article is as follows. Section 2 introduces notations and preliminaries. Section 3 defines SVDinsTN, constructs an energy function, develops the corresponding optimization method, and conducts experiments to show the effectiveness of SVDinsTN. Section 4 applies SVDinsTN to real-world applications and compares

SVDinsTN with SOTA tensor methods. Section 5 discusses the impact of different sparsity-inducing functions and key parameter settings. Section 6 concludes the article.

2 NOTATIONS AND PRELIMINARIES

A *tensor* is a multidimensional array, and the number of dimensions, also called *modes*, is referred to as the *tensor order*. We use bold lowercase letters, e.g., $\mathbf{x}$, bold uppercase letters, e.g., $\mathbf{X}$, and calligraphic uppercase letters, e.g., $\mathcal{X}$, to denote first-order tensors (vectors), second-order tensors (matrices), and N th-order tensors ($N \geq 3$), respectively. For an N th-order tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$, its Frobenius norm is $\|\mathcal{X}\|_F = \sqrt{\sum_{i_1, i_2, \dots, i_N} |\mathcal{X}(i_1, i_2, \dots, i_N)|^2}$ and its ℓ_1 -norm is $\|\mathcal{X}\|_1 = \sum_{i_1, i_2, \dots, i_N} |\mathcal{X}(i_1, i_2, \dots, i_N)|$. To simplify notation, we let $\mathcal{X}_{1:d}$ denote the ordered set $\{\mathcal{X}_1, \mathcal{X}_2, \dots, \mathcal{X}_d\}$, $\mathbb{K}_N$ denote the set $\{1, 2, \dots, N\}$, and $\mathbb{TL}_N$ denote the set $\{(t, l) | 1 \leq t < l \leq N; t, l \in \mathbb{N}\}$. We use $\pi(N)$ to denote the set of all possible rearrangements of the sequence $(1, 2, \dots, N)$.

We next review several operations on tensors [38].

Definition 1 (Transposition). Let $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ and $\mathbf{n} = (n_1, n_2, \dots, n_N) \in \pi(N)$. The $\mathbf{n}$ -based transposition of $\mathcal{X}$ is obtained by permuting its modes according to the order $\mathbf{n}$, and is denoted by $\mathcal{X}^{\mathbf{n}} \in \mathbb{R}^{I_{n_1} \times I_{n_2} \times \dots \times I_{n_N}}$. This operation is written as $\mathcal{X}^{\mathbf{n}} = \text{transpose}(\mathcal{X}, \mathbf{n})$, and its inverse is written as $\mathcal{X} = \text{itranspose}(\mathcal{X}^{\mathbf{n}}, \mathbf{n})$.

Definition 2 (Unfolding). Let $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$, $\mathbf{n} \in \pi(N)$, and $d \in \{1, 2, \dots, N-1\}$. The $\prod_{i=1}^d I_{n_i} \times \prod_{i=d+1}^N I_{n_i}$ matrix $\mathbf{X}_{[n_{1:d}|n_{d+1:N}]}$ satisfying $\mathbf{X}_{[n_{1:d}|n_{d+1:N}]}(j_1, j_2) = \mathcal{X}(i_1, i_2, \dots, i_N)$ with

$$\begin{cases} j_1 = i_{n_1} + \sum_{s=2}^d \left((i_{n_s} - 1) \prod_{m=1}^{s-1} I_{n_m} \right), \\ j_2 = i_{n_{d+1}} + \sum_{s=d+2}^N \left((i_{n_s} - 1) \prod_{m=d+1}^{s-1} I_{n_m} \right), \end{cases}$$

is called the $(\mathbf{n}, d)$ -based unfolding of $\mathcal{X}$. The corresponding operation and its inverse are written as $\mathbf{X}_{[n_{1:d}|n_{d+1:N}]} = \text{GenUnfold}(\mathcal{X}, \mathbf{n}_{1:d}|n_{d+1:N})$ and $\mathcal{X} = \text{GenFold}(\mathbf{X}_{[n_{1:d}|n_{d+1:N}]}, n_{1:d}|n_{d+1:N})$, respectively.

We also use $\mathbf{X}_{(k)}$ to denote $\mathbf{X}_{[k|1,2,\dots,k-1,k+1,\dots,N]}$, since it is the traditional mode- k unfolding of $\mathcal{X}$ [10]. Besides, we use $\mathbf{X}_{(k)} = \text{Unfold}(\mathcal{X}, k)$ and $\mathcal{X} = \text{Fold}(\mathbf{X}_{(k)}, k)$ to simply denote the traditional mode- k unfolding operation and its inverse operation, respectively.

Definition 3 (Tensor Contraction). Let $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ and $\mathcal{Y} \in \mathbb{R}^{J_1 \times J_2 \times \dots \times J_M}$ satisfy $I_{n_i} = J_{m_i}$ for $i = 1, 2, \dots, d$ and $d \in \{1, 2, \dots, \min\{M, N\}\}$. Here $\mathbf{n} \in \pi(N)$ and $\mathbf{m} \in \pi(M)$ satisfy that $n_{d+1} < n_{d+2} < \dots < n_N$ and $m_{d+1} < m_{d+2} < \dots < m_M$, respectively. The tensor contraction between the modes indexed by $n_{1:d}$ of $\mathcal{X}$ and the modes indexed by $m_{1:d}$ of $\mathcal{Y}$ yields an $(N + M - 2d)$ -th-order tensor

$$\mathcal{Z} = \mathcal{X} \times_{n_{1:d}}^{m_{1:d}} \mathcal{Y} \in \mathbb{R}^{I_{n_{d+1}} \times \dots \times I_{n_N} \times J_{m_{d+1}} \times \dots \times J_{m_M}},$$

whose entries are formulated as

$$\mathcal{Z}(i_{n_{d+1}}, \dots, i_{n_N}, j_{m_{d+1}}, \dots, j_{m_M}) = \sum_{i_{n_1}=1}^{I_{n_1}} \sum_{i_{n_2}=1}^{I_{n_2}} \dots \sum_{i_{n_d}=1}^{I_{n_d}} \{ \mathcal{X}^{\mathbf{n}}(i_{n_1}, \dots, i_{n_d}, i_{n_{d+1}}, \dots, i_{n_N}) \mathcal{Y}^{\mathbf{m}}(i_{n_1}, \dots, i_{n_d}, j_{m_{d+1}}, \dots, j_{m_M}) \}.$$

The operation $\text{diag}(\mathbf{X})$ extracts the diagonal entries of a matrix $\mathbf{X}$ and returns them as a column vector. Conversely, when the input is a column vector $\mathbf{x}$, $\text{diag}(\mathbf{x})$ constructs a diagonal matrix with the entries of $\mathbf{x}$ along its main diagonal. The function $\text{ones}(I_1, I_2, \dots, I_N)$ creates a tensor of size $I_1 \times I_2 \times \dots \times I_N$, with every element set to 1. The operation $\text{vec}(\mathcal{X})$ transforms the tensor $\mathcal{X}$ into a column vector by reordering its entries in lexicographical order.

2.1 Tensor Network Decomposition

A tensor network (TN) decomposition represents a large-scale higher-order tensor through a set of smaller tensors, referred to as *TN cores*. These cores are contracted along some or all modes according to specific rules [34]. This article focuses on a particular type of TN decomposition, where TN cores are connected through tensor contractions and the TN topology is an unweighted simple graph. Additionally, in the representation of an N th-order tensor $\mathcal{X}$, this type of TN decomposition uses exactly N TN cores, each corresponding to a different mode of $\mathcal{X}$. In this type of TN decomposition, the choice of TN topology leads to distinct decomposition formats. Notable examples of such formats include the TT decomposition [36], TR decomposition [37], and FCTN decomposition [38].

TT Decomposition [36] represents an N th-order tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ as

$$\mathcal{X}(i_1, i_2, \dots, i_N) = \sum_{r_{1,2}=1}^{R_{1,2}} \sum_{r_{2,3}=1}^{R_{2,3}} \dots \sum_{r_{N-1,N}=1}^{R_{N-1,N}} \quad (1)$$

$$\mathbf{G}_1(i_1, r_{1,2}) \mathcal{G}_2(r_{1,2}, i_2, r_{2,3}) \dots \mathcal{G}_{N-1}(r_{N-2,N-1}, i_{N-1}, r_{N-1,N}) \mathbf{G}_N(r_{N-1,N}, i_N),$$

where $\mathbf{G}_1 \in \mathbb{R}^{I_1 \times R_{1,2}}$, $\mathcal{G}_k \in \mathbb{R}^{R_{k-1,k} \times I_k \times R_{k,k+1}}$ for $k = 2, 3, \dots, N-1$, and $\mathbf{G}_N \in \mathbb{R}^{R_{N-1,N} \times I_N}$ are TT cores. We denote (1) by $\mathcal{X} = \text{TT}(\mathbf{G}_1, \{\mathcal{G}_k\}_{k=2}^{N-1}, \mathbf{G}_N)$ and define the TT rank as $\mathbf{r}_{\text{TT}} = (R_{1,2}, R_{2,3}, \dots, R_{N-1,N}) \in \mathbb{R}^{N-1}$.

TR Decomposition [37] represents an N th-order tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ as

$$\mathcal{X}(i_1, i_2, \dots, i_N) = \sum_{r_{1,2}=1}^{R_{1,2}} \sum_{r_{2,3}=1}^{R_{2,3}} \dots \sum_{r_{N-1,N}=1}^{R_{N-1,N}} \sum_{r_{1,N}=1}^{R_{1,N}} \quad (2)$$

$$\mathcal{G}_1(r_{1,N}, i_1, r_{1,2}) \mathcal{G}_2(r_{1,2}, i_2, r_{2,3}) \dots \mathcal{G}_{N-1}(r_{N-2,N-1}, i_{N-1}, r_{N-1,N}) \mathcal{G}_N(r_{N-1,N}, i_N, r_{1,N}),$$

where $\mathcal{G}_1 \in \mathbb{R}^{R_{1,N} \times I_1 \times R_{1,2}}$, $\mathcal{G}_k \in \mathbb{R}^{R_{k-1,k} \times I_k \times R_{k,k+1}}$ for $k = 2, 3, \dots, N-1$, and $\mathcal{G}_N \in \mathbb{R}^{R_{N-1,N} \times I_N \times R_{1,N}}$ are TR cores. We denote (2) by $\mathcal{X} = \text{TR}(\{\mathcal{G}_k\}_{k=1}^N)$ and define the TR rank as $\mathbf{r}_{\text{TR}} = (R_{1,2}, R_{2,3}, \dots, R_{1,N}) \in \mathbb{R}^N$.

Algorithm 1 $\mathcal{X} = \text{FCTN}(\{\mathcal{G}_k\}_{k=1}^N)$.

Input: $\mathcal{G}_k \in \mathbb{R}^{R_{1,k} \times \dots \times R_{k-1,k} \times I_k \times R_{k,k+1} \times \dots \times R_{k,N}}$ for all $k \in \mathbb{K}_N$.
 $a_1 = 1, b_1 = 2$, and $\mathcal{X} = \mathcal{G}_1$.
for $i = 1$ to $N - 1$ **do**
 $\mathcal{X} = \mathcal{X} \times_{b_1, b_2, \dots, b_i}^{a_1, a_2, \dots, a_i} \mathcal{G}_{i+1}$.
for $j = 1$ to $i + 1$ **do**
 $a_j = j$.
 $b_j = 2 + (j - 1)(N - i)$.
end for
end for
Output: $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$.

FCTN Decomposition [38] represents an N th-order tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ as

$$\begin{aligned} \mathcal{X}(i_1, i_2, \dots, i_N) = & \sum_{r_{1,2}=1}^{R_{1,2}} \sum_{r_{1,3}=1}^{R_{1,3}} \dots \sum_{r_{1,N}=1}^{R_{1,N}} \sum_{r_{2,3}=1}^{R_{2,3}} \sum_{r_{2,N}=1}^{R_{2,N}} \dots \sum_{r_{N-1,N}=1}^{R_{N-1,N}} \\ & \mathcal{G}_1(i_1, r_{1,2}, r_{1,3}, \dots, r_{1,N}) \\ & \mathcal{G}_2(r_{1,2}, i_2, r_{2,3}, \dots, r_{2,N}) \dots \\ & \mathcal{G}_k(r_{1,k}, r_{2,k}, \dots, r_{k-1,k}, i_k, r_{k,k+1}, \dots, r_{k,N}) \dots \\ & \mathcal{G}_N(r_{1,N}, r_{2,N}, \dots, r_{N-1,N}, i_N), \end{aligned} \quad (3)$$

where $\mathcal{G}_k \in \mathbb{R}^{R_{1,k} \times \dots \times R_{k-1,k} \times I_k \times R_{k,k+1} \times \dots \times R_{k,N}}$ for all $k \in \mathbb{K}_N$ are N th-order tensors, called FCTN cores. We denote (3) by $\mathcal{X} = \text{FCTN}(\{\mathcal{G}_k\}_{k=1}^N)$ and define the FCTN rank as $\mathbf{r}_{\text{FCTN}} = (R_{1,2}, R_{1,3}, \dots, R_{1,N}, R_{2,3}, \dots, R_{2,N}, \dots, R_{N-1,N}) \in \mathbb{R}^{\frac{N(N-1)}{2}}$. Algorithm 1 gives a way to compute (3).

Fig. 1 illustrates the structures of TT, TR, and FCTN decompositions. TT and TR represent an N th-order tensor using low-order cores connected only to their neighboring cores. This local connectivity improves storage and computational efficiency, but may limit their ability to model global correlations among tensor modes. In contrast, FCTN allows pairwise contractions among all TN cores, which provides stronger flexibility for capturing global dependencies. However, the transition from “chain” to “ring” and then to “fully-connected” topology also increases the complexity of TN decompositions. As a result, more rank parameters need to be determined, making the deployment of TN decompositions more challenging in practical applications.

Remark 1. The TN decompositions considered in this article can be regarded as special instances of FCTN decomposition. In a TN structure, an edge with rank one does not introduce an effective interaction between two cores and thus can be removed without changing the represented tensor. Therefore, when an element of the FCTN rank equals one, the corresponding edge in the “fully-connected topology” can be omitted. For example, if the FCTN rank is $(R_{1,2}, 1, \dots, 1, R_{2,3}, 1, \dots, R_{N-1,N})$, the “fully-connected” topology can be seamlessly transformed into a “chain” topology (TT decomposition), and the TT rank is $(R_{1,2}, R_{2,3}, \dots, R_{N-1,N})$.

The statement in Remark 1 can be rigorously described in the following formal statement.

Property 1. [58] There is a one-to-one relationship between the FCTN rank and the TN structure.

Based on Property 1, we can transform the TN structure search problem into an FCTN rank optimization problem.

3 SVD-INSPIRED TN DECOMPOSITION

In this section, we define SVDinsTN and present its related theories. We then construct an energy function and develop the corresponding optimization method to reveal the TN structure concurrently with the computation of the TN cores. Finally, we perform experiments to validate the effectiveness of SVDinsTN.

3.1 SVDinsTN

The definition of SVDinsTN is given as follows.

Definition 4 (SVD-inspired TN decomposition, SVDinsTN). Let $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ be an N th-order tensor that can be represented as

$$\begin{aligned} \mathcal{X}(i_1, i_2, \dots, i_N) = & \sum_{r_{1,2}=1}^{R_{1,2}} \sum_{r_{1,3}=1}^{R_{1,3}} \dots \sum_{r_{1,N}=1}^{R_{1,N}} \sum_{r_{2,3}=1}^{R_{2,3}} \sum_{r_{2,N}=1}^{R_{2,N}} \dots \sum_{r_{N-1,N}=1}^{R_{N-1,N}} \\ & \mathbf{S}_{1,2}(r_{1,2}, r_{1,2}) \mathbf{S}_{1,3}(r_{1,3}, r_{1,3}) \dots \mathbf{S}_{1,N}(r_{1,N}, r_{1,N}) \\ & \mathbf{S}_{2,3}(r_{2,3}, r_{2,3}) \dots \mathbf{S}_{2,N}(r_{2,N}, r_{2,N}) \dots \\ & \mathbf{S}_{N-1,N}(r_{N-1,N}, r_{N-1,N}) \\ & \mathcal{G}_1(i_1, r_{1,2}, r_{1,3}, \dots, r_{1,N}) \\ & \mathcal{G}_2(r_{1,2}, i_2, r_{2,3}, \dots, r_{2,N}) \dots \\ & \mathcal{G}_k(r_{1,k}, r_{2,k}, \dots, r_{k-1,k}, i_k, r_{k,k+1}, \dots, r_{k,N}) \dots \\ & \mathcal{G}_N(r_{1,N}, r_{2,N}, \dots, r_{N-1,N}, i_N), \end{aligned} \quad (4)$$

where $\mathbf{S}_{t,l} \in \mathbb{R}^{R_{t,l} \times R_{t,l}}$ are diagonal matrices for $(t, l) \in \mathbb{TL}_N$, and $\mathcal{G}_k \in \mathbb{R}^{R_{1,k} \times \dots \times R_{k-1,k} \times I_k \times R_{k,k+1} \times \dots \times R_{k,N}}$ are N th-order tensors for $k \in \mathbb{K}_N$, which are called TN cores. We simply denote (4) by $\mathcal{X} = \text{STN}(\mathcal{G}, \mathbf{S})$ and call it an SVDinsTN of $\mathcal{X}$, where $\mathcal{G}$ represents $\{\mathcal{G}_k | k \in \mathbb{K}_N\}$ and $\mathbf{S}$ represents $\{\mathbf{S}_{t,l} | (t, l) \in \mathbb{TL}_N\}$.

The following theorem explains how SVDinsTN effectively reveals the TN structure.

Theorem 1. SVDinsTN can be transformed into a TN decomposition, where the numbers of non-zero diagonal entries in the diagonal matrices determine the TN structure, including the TN topology and TN rank.

Proof. Assume that $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ is an N th-order tensor and can be expressed as $\mathcal{X} = \text{STN}(\mathcal{G}, \mathbf{S})$, where $\mathcal{G}$ and $\mathbf{S}$ are defined as in Definition 4. We let

$$\begin{cases} \bar{\mathcal{G}}_1 = \mathcal{G}_1 \times_2^1 \mathbf{S}_{1,2} \times_2^1 \mathbf{S}_{1,3} \times_2^1 \dots \times_2^1 \mathbf{S}_{1,N} \\ \bar{\mathcal{G}}_2 = \mathcal{G}_2 \times_3^1 \mathbf{S}_{2,3} \times_3^1 \mathbf{S}_{2,4} \times_3^1 \dots \times_3^1 \mathbf{S}_{2,N} \\ \vdots \\ \bar{\mathcal{G}}_{N-1} = \mathcal{G}_{N-1} \times_N^1 \mathbf{S}_{N-1,N} \\ \bar{\mathcal{G}}_N = \mathcal{G}_N \end{cases}, \quad (5)$$

then we have $\mathcal{X} = \text{FCTN}(\{\bar{\mathcal{G}}_k\}_{k=1}^N)$. Without loss of generality, we assume that diagonal matrices $\mathbf{S}_{t,l} \in \mathbb{R}^{R_{t,l} \times R_{t,l}}$ for all $(t, l) \in \mathbb{TL}_N$ have their first $\bar{R}_{t,l}$ diagonal entries as non-zero, with all other entries being zero.

Algorithm 2 Generate matrix $\mathbf{M}_k$ in Theorem 2.

Input: $k \in \mathbb{K}_N$, $\mathcal{G}_{k'} \in \mathbb{R}^{R_{1,k'} \times \dots \times R_{k'-1,k'} \times I_{k'} \times R_{k',k'+1} \times \dots \times R_{k',N}}$ for all $k' \in \mathbb{K}_N$ with $k' \neq k$; $\mathbf{S}_{t,l} \in \mathbb{R}^{R_{t,l} \times R_{t,l}}$ for all $(t,l) \in \mathbb{TL}_N$.
Initialization: $\mathbf{a} = (1:k-1, N, k:N-1)$ and $\mathbf{b} = (k+1:N, 1:k)$.
for $t = 1$ to N and $t \neq k$ **do**
 for $l = t+1$ to N **do**
 $\mathcal{G}_t = \mathcal{G}_t \times_{t+1}^1 \mathbf{S}_{t,l}$.
 end for
 if $t > k$ **then**
 $\mathcal{G}_t = \mathcal{G}_t \times_k^1 \mathbf{S}_{k,t}$ and $\mathcal{G}_t = \text{transpose}(\mathcal{G}_t, \mathbf{a})$.
 end if
 $\mathcal{G}_t = \text{transpose}(\mathcal{G}_t, \mathbf{b})$.
end for
 $c_1 = 1, d_1 = 2$, and $\mathcal{M}_k = \mathcal{G}_{\mathbf{b}(1)}$.
for $i = 1$ to $N-2$ **do**
 $\mathcal{M}_k = \mathcal{M}_k \times_{d_1, d_2, \dots, d_i}^{c_1, c_2, \dots, c_i} \mathcal{G}_{\mathbf{b}(i+1)}$.
 for $j = 1$ to $i+1$ **do**
 $c_j = j$.
 $d_j = 2 + (j-1)(N-i)$.
 end for
end for
 $\mathcal{M}_k = \text{transpose}(\mathcal{M}_k, (2(N-k)+1:2(N-1), 1:2(N-k)))$.
 $\mathbf{e} = \text{ones}(1, N-1)$ and $\mathbf{f} = \text{ones}(1, N-1)$.
for $i = 1$ to $N-1$ **do**
 $\mathbf{e}(i) = 2i$ and $\mathbf{f}(i) = 2i-1$.
end for
 $\mathbf{M}_k = \text{GenUnfold}(\mathcal{M}_k, (\mathbf{e}|\mathbf{f}))$.
Output: $\mathbf{M}_k \in \mathbb{R}^{\prod_{i=1}^{k-1} R_{i,k} \times \prod_{i=k+1}^N R_{k,i} \times \prod_{i=1, i \neq k}^N I_i}$.

Then, for each (t, l) , the terms associated with $r_{t,l} > \bar{R}_{t,l}$ make no contribution to the final tensor contraction. Therefore, the summation range of each index $r_{t,l}$ can be restricted from $\{1, 2, \dots, R_{t,l}\}$ to $\{1, 2, \dots, \bar{R}_{t,l}\}$. Let

$$\bar{\mathcal{G}}'_k = \bar{\mathcal{G}}_k(1:\bar{R}_{1,k}, \dots, 1:\bar{R}_{k-1,k}, :, 1:\bar{R}_{k,k+1}, \dots, 1:\bar{R}_{k,N})$$

for all $k \in \mathbb{K}_N$, we can obtain that $\mathcal{X} = \text{FCTN}(\{\bar{\mathcal{G}}'_k\}_{k=1}^N)$, where $\bar{\mathcal{G}}'_k \in \mathbb{R}^{\bar{R}_{1,k} \times \dots \times \bar{R}_{k-1,k} \times I_k \times \bar{R}_{k,k+1} \times \dots \times \bar{R}_{k,N}}$. Therefore, SVDinsTN can be transformed into a FCTN decomposition and the number of non-zero entries in the diagonal matrices determines the FCTN rank. According to Remark 1 and Property 1, Theorem 1 holds. $\square$

Theorem 1 shows that SVDinsTN can identify a compact TN structure through the sparsity of the inserted diagonal matrices, while representing the given tensor using the TN cores with these matrices incorporated. Therefore, we can construct an energy function that jointly accounts for the representation error and the complexity of the TN structure. This formulation makes the model amenable to efficient optimization with well-established algorithms. Before presenting the method, we first introduce several theorems that serve as its theoretical basis.

(1) *Computing $\mathcal{X} = \text{STN}(\mathcal{G}, \mathbf{S})$ with given $\mathcal{G}$ and $\mathbf{S}$.* In fact, Theorem 1 provides a practical way to compute $\mathcal{X} = \text{STN}(\mathcal{G}, \mathbf{S})$. Specifically, we can first generate $\bar{\mathcal{G}}$ using (5) and then compute $\mathcal{X} = \text{FCTN}(\{\bar{\mathcal{G}}'_k\}_{k=1}^N)$ through Algorithm 1.

(2) *Relationship between each TN core or diagonal matrix and the remaining components.* This relationship is essential for the alternating optimization during the solution process, since it enables us to iteratively refine each core or matrix by fixing the others. This relationship is characterized by the following two theorems.

Theorem 2. Let $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ be an N th-order tensor

Algorithm 3 Generate matrix $\mathbf{H}_{t,l}$ in Theorem 3.

Input: $(t, l) \in \mathbb{TL}_N$, $\mathbf{S}_{t',l'} \in \mathbb{R}^{R_{t',l'} \times R_{t',l'}}$ for all $(t', l') \in \mathbb{TL}_N$ with $(t', l') \neq (t, l)$; $\mathcal{G}_k \in \mathbb{R}^{R_{1,k} \times \dots \times R_{k-1,k} \times I_k \times R_{k,k+1} \times \dots \times R_{k,N}}$ for all $k \in \mathbb{K}_N$.
for $t' = 1$ to $N-1$ and $t' \neq t$ **do**
 for $l' = t'+1$ to N **do**
 $\mathcal{G}_{t'} = \mathcal{G}_{t'} \times_{t'+1}^1 \mathbf{S}_{t',l'}$.
 end for
end for
for $l' = t+1$ to N and $l' \neq l$ **do**
 if $l' < l$ **then**
 $\mathcal{G}_t = \mathcal{G}_t \times_{t+1}^1 \mathbf{S}_{t,l'}$.
 else
 $\mathcal{G}_t = \mathcal{G}_t \times_{t+2}^1 \mathbf{S}_{t,l'}$.
 end if
end for
 $\mathcal{G}_t = \text{transpose}(\mathcal{G}_t, (1:t, t+2:l, t+1, l+1:N))$.
 $\mathbf{H}_{t,l} = \text{ones}(\prod_{k=1}^N I_k, R_{t,l})$.
 $\mathbf{G}_t = \text{Unfold}(\mathcal{G}_t, l)$ and $\mathbf{G}_l = \text{Unfold}(\mathcal{G}_l, t)$.
for $i = 1$ to $R_{t,l}$ **do**
 $\mathcal{G}_t = \text{Fold}(\mathbf{G}_t(i,:), l)$ and $\mathcal{G}_l = \text{Fold}(\mathbf{G}_l(i,:), t)$.
 $\mathbf{H}_{t,l}(:, i) = \text{vec}(\text{FCTN}(\{\mathcal{G}_k\}_{k=1}^N))$.
end for
Output: $\mathbf{H}_{t,l} \in \mathbb{R}^{\prod_{k=1}^N I_k \times R_{t,l}}$.

expressed as the SVDinsTN in (4), where $\mathcal{G}_k$ is a TN core for $k \in \mathbb{K}_N$, and $\mathbf{M}_k$ is the composition of all TN cores and diagonal matrices except $\mathcal{G}_k$, computed by Algorithm 2. Then, the following equation holds:

$$\mathbf{X}_{(k)} = \mathbf{G}_{(k)} \mathbf{M}_k.$$

Theorem 3. Let $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ be an N th-order tensor expressed as the SVDinsTN in (4), where $\mathbf{S}_{t,l}$ is a diagonal matrix for $(t, l) \in \mathbb{TL}_N$, and $\mathbf{H}_{t,l}$ is the composition of all TN cores and diagonal matrices except $\mathbf{S}_{t,l}$, computed by Algorithm 3. Then, the following equation holds:

$$\mathbf{x} = \mathbf{H}_{t,l} \mathbf{s}_{t,l},$$

where $\mathbf{x} = \text{vec}(\mathcal{X})$ and $\mathbf{s}_{t,l} = \text{diag}(\mathbf{S}_{t,l})$.

It is worth noting that Theorems 2 and 3 also allow us to transform the complex tensor equation into a matrix equation, simplifying the optimization process. The following remark provides a brief third-order case to illustrate the basic ideas behind the generation of $\mathbf{M}_k$ and $\mathbf{H}_{t,l}$.

Remark 2. For example, consider a third-order tensor. To generate $\mathbf{M}_3$, Algorithm 2 incorporates the associated diagonal matrices into the TN cores except $\mathcal{G}_3$. The processed cores are then contracted along the shared rank dimension and unfolded to obtain $\mathbf{M}_3$. Similarly, to generate $\mathbf{H}_{2,3}$, Algorithm 3 preserves $\mathbf{S}_{2,3}$ and incorporates the remaining diagonal matrices into the corresponding TN cores. Then, $\mathcal{G}_2$ and $\mathcal{G}_3$, which share the rank- $R_{2,3}$ mode, are unfolded along this rank dimension. Each rank component is subsequently contracted with the remaining core and vectorized to form the corresponding column of $\mathbf{H}_{2,3}$.

(3) *An upper bound for the TN rank (structure).* This upper bound provides valuable guidance for designing an efficient initialization scheme in practical applications. This is formalized in the following theorem.

Theorem 4. Let $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ be a non-zero N th-order tensor. Then there exists an SVDinsTN (4) with the rank $R_{t,l} \leq \min(\text{rank}(\mathbf{X}_{(t)}), \text{rank}(\mathbf{X}_{(l)}))$ for all $(t, l) \in \mathbb{T}\mathbb{L}_N$.

Before proving Theorem 4, we first establish a lemma that plays a pivotal role in the proof of Theorem 4.

Lemma 1. Let $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ be a non-zero N th-order tensor. Then there exists an SVDinsTN with the rank $R_{t,l} \leq \min(I_t, I_l)$ for all $(t, l) \in \mathbb{T}\mathbb{L}_N$.

Proof. Without loss of generality, we assume that $I_1 \geq I_2 \geq \dots \geq I_N$, and set $R_{t,l} = \min(I_t, I_l) = I_l$ for all $(t, l) \in \mathbb{T}\mathbb{L}_N$. Let $\mathcal{G}_k \in \mathbb{R}^{R_{1,k} \times R_{2,k} \times \dots \times R_{k-1,k} \times I_k \times R_{k,k+1} \times \dots \times R_{k,N}}$ for all $k \in \mathbb{K}_N$ satisfy $\mathcal{G}_1 = \mathcal{X}$ and

$$\mathcal{G}_k(r_{1,k}, r_{2,k}, \dots, r_{k-1,k}, i_k, r_{k,k+1}, \dots, r_{k,N}) = \begin{cases} 1, & \text{if } r_{1,k} = i_k, r_{m,k} = 1 \ (m = 2, \dots, k-1), \\ & \text{and } r_{k,n} = 1 \ (n = k+1, \dots, N), \\ 0, & \text{otherwise,} \end{cases}$$

for $k = 2, 3, \dots, N$. Also, let $\mathbf{S}_{t,l} \in \mathbb{R}^{R_{t,l} \times R_{t,l}}$ for all $(t, l) \in \mathbb{T}\mathbb{L}_N$ be identity matrices. Then we have $\mathcal{X} = \text{STN}(\mathcal{G}, \mathbf{S})$. Thus, there exists an SVDinsTN with $R_{t,l} \leq \min(I_t, I_l)$ for all $(t, l) \in \mathbb{T}\mathbb{L}_N$. Therefore, Lemma 1 holds. $\square$

Below, we prove Theorem 4, which asserts that for an N th-order tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$, the Tucker rank $(r_1, r_2, \dots, r_N) = (\text{rank}(\mathbf{X}_{(1)}), \text{rank}(\mathbf{X}_{(2)}), \dots, \text{rank}(\mathbf{X}_{(N)}))$ serves as an upper bound for the TN rank.

Proof. Without loss of generality, we assume that the Tucker rank of $\mathcal{X}$ satisfies $r_1 \geq r_2 \geq \dots \geq r_N$. Then, there exists a Tucker decomposition as follows:

$$\mathcal{X} = \mathcal{C} \times_1 \mathbf{U}_1 \times_2 \mathbf{U}_2 \times_3 \dots \times_N \mathbf{U}_N,$$

where $\mathcal{C} \in \mathbb{R}^{r_1 \times r_2 \times \dots \times r_N}$, $\mathbf{U}_k \in \mathbb{R}^{I_k \times r_k}$ for $k = 1, 2, \dots, N$, and $\times_k$ is the k -mode product. According to Lemma 1, for the tensor $\mathcal{C}$, there exists an SVDinsTN with $R_{t,l} = r_l$ for all $(t, l) \in \mathbb{T}\mathbb{L}_N$. That is, $\mathcal{C} = \text{STN}(\bar{\mathcal{G}}, \mathbf{S})$, where $\bar{\mathcal{G}}$ represents $\{\bar{\mathcal{G}}_k | k \in \mathbb{K}_N\}$, $\bar{\mathcal{G}}_k \in \mathbb{R}^{r_k \times \dots \times r_k \times r_k \times r_{k+1} \times r_{k+2} \times \dots \times r_N}$ for all $k \in \mathbb{K}_N$, and $\mathbf{S}_{t,l} \in \mathbb{R}^{r_l \times r_l}$ for all $(t, l) \in \mathbb{T}\mathbb{L}_N$. Let

$$\mathcal{G}_k = \bar{\mathcal{G}}_k \times_k \mathbf{U}_k \text{ for all } k \in \mathbb{K}_N.$$

Then we have $\mathcal{X} = \text{STN}(\mathcal{G}, \mathbf{S})$. That is, the SVDinsTN of $\mathcal{X}$ always exists with $R_{t,l} = r_l$ for all $(t, l) \in \mathbb{T}\mathbb{L}_N$. Therefore, Theorem 4 holds. $\square$

3.2 An Energy Function for SVDinsTN

Based on SVDinsTN, we construct an energy function that accounts for both representation error and the complexity of the TN structure simultaneously. Mathematically, the energy function is defined as

$$f(\mathcal{G}, \mathbf{S}) = \frac{1}{2} \|\mathcal{X} - \text{STN}(\mathcal{G}, \mathbf{S})\|_F^2 + \lambda \phi(\mathbf{S}) + \mu \varphi(\mathcal{G}), \quad (6)$$

where $\mathcal{X}$ is a given tensor, $\lambda, \mu > 0$ are balance parameters, $\|\mathcal{X} - \text{STN}(\mathcal{G}, \mathbf{S})\|_F^2$ is used to measure the representation error, $\phi(\cdot)$ is a sparsity-inducing term used to capture the complexity of the TN structure, and $\varphi(\cdot)$ is a constraint used to bound the feasible range of $\mathcal{G}$. This energy function allows us to leverage various sparsity-inducing terms, such as ℓ_0 -norm, ℓ_1 -norm, and log-norm, to learn the number of non-zero entries in $\mathbf{S}$. Additionally, it provides the flexibility

to impose different constraints on $\mathcal{G}$ to constrain the feasible range of $\mathcal{G}$. By minimizing the energy function, the TN cores and diagonal matrices in SVDinsTN are updated iteratively, with the sparsity of the diagonal matrices revealing a compact TN structure.

Below, we consider the basic ℓ_1 -norm for diagonal matrices and the squared Frobenius norm for TN cores as an illustrative example. A more detailed discussion of other sparsity-inducing terms can be found in Section 5. The ℓ_1 -norm-based energy model can be formulated as

$$\min_{\mathcal{G}, \mathbf{S}} \frac{1}{2} \|\mathcal{X} - \text{STN}(\mathcal{G}, \mathbf{S})\|_F^2 + \frac{\mu}{2} \sum_{k \in \mathbb{K}_N} \|\mathcal{G}_k\|_F^2 + \sum_{(t,l) \in \mathbb{T}\mathbb{L}_N} \lambda_{t,l} \|\mathbf{S}_{t,l}\|_1, \quad (7)$$

where $\lambda_{t,l} > 0$ controls the sparsity of diagonal matrices and $\mu > 0$ balances the regularization of TN cores.

We employ the basic PAM [59] to solve (7), which alternately updates the TN cores and diagonal matrices with a proximal term incorporated into each block subproblem to enhance optimization stability. Under this framework, the update subproblems are formulated as follows:

$$\begin{cases} \mathcal{G}_k = \underset{\mathcal{G}_k}{\text{argmin}} \frac{1}{2} \|\mathcal{X} - \text{STN}(\mathcal{G}, \mathbf{S})\|_F^2 + \frac{\mu}{2} \|\mathcal{G}_k\|_F^2 \\ \quad + \frac{\rho}{2} \|\mathcal{G}_k - \hat{\mathcal{G}}_k\|_F^2, \quad \forall k \in \mathbb{K}_N, \\ \mathbf{S}_{t,l} = \underset{\mathbf{S}_{t,l}}{\text{argmin}} \frac{1}{2} \|\mathcal{X} - \text{STN}(\mathcal{G}, \mathbf{S})\|_F^2 + \lambda_{t,l} \|\mathbf{S}_{t,l}\|_1 \\ \quad + \frac{\rho}{2} \|\mathbf{S}_{t,l} - \hat{\mathbf{S}}_{t,l}\|_F^2, \quad \forall (t, l) \in \mathbb{T}\mathbb{L}_N, \end{cases} \quad (8)$$

where $\rho > 0$ is a proximal parameter, and $\hat{\mathcal{G}}_k$ and $\hat{\mathbf{S}}_{t,l}$ denote the corresponding solutions from the previous iteration.

1) *Update $\mathcal{G}_k$ for $k \in \mathbb{K}_N$:* Based on Theorem 2, the $\mathcal{G}_k$ -subproblem can be reformulated as

$$\min_{\mathbf{G}_{k(k)}} \frac{1}{2} \|\mathbf{X}_{(k)} - \mathbf{G}_{k(k)} \mathbf{M}_k\|_F^2 + \frac{\mu}{2} \|\mathbf{G}_{k(k)}\|_F^2 + \frac{\rho}{2} \|\mathbf{G}_{k(k)} - \hat{\mathbf{G}}_{k(k)}\|_F^2, \quad (9)$$

whose solution is

$$\mathbf{G}_{k(k)} = (\mathbf{X}_{(k)} \mathbf{M}_k^T + \rho \hat{\mathbf{G}}_{k(k)}) (\mathbf{M}_k \mathbf{M}_k^T + (\mu + \rho) \mathbf{I})^{-1}. \quad (10)$$

2) *Update $\mathbf{S}_{t,l}$ for $(t, l) \in \mathbb{T}\mathbb{L}_N$:* Based on Theorem 3, the $\mathbf{S}_{t,l}$ -subproblem can be reformulated as

$$\min_{\mathbf{s}_{t,l}} \frac{1}{2} \|\mathbf{x} - \mathbf{H}_{t,l} \mathbf{s}_{t,l}\|_F^2 + \lambda_{t,l} \|\mathbf{s}_{t,l}\|_1 + \frac{\rho}{2} \|\mathbf{s}_{t,l} - \hat{\mathbf{s}}_{t,l}\|_F^2. \quad (11)$$

We use ADMM [60] to solve the $\mathbf{S}_{t,l}$ -subproblem, which can be reformulated as

$$\min_{\mathbf{s}_{t,l}, \mathbf{q}_{t,l}} \frac{1}{2} \|\mathbf{x} - \mathbf{H}_{t,l} \mathbf{q}_{t,l}\|_F^2 + \lambda_{t,l} \|\mathbf{s}_{t,l}\|_1 + \frac{\rho}{2} \|\mathbf{s}_{t,l} - \hat{\mathbf{s}}_{t,l}\|_F^2 \quad (12)$$

$$\text{s.t. } \mathbf{s}_{t,l} - \mathbf{q}_{t,l} = 0,$$

where $\mathbf{q}_{t,l}$ is an auxiliary variable introduced to facilitate the solution. We first formulate the augmented Lagrangian function for the problem in (12) as

$$L_{\beta_{t,l}}(\mathbf{s}_{t,l}, \mathbf{q}_{t,l}, \mathbf{p}_{t,l}) = \frac{1}{2} \|\mathbf{x} - \mathbf{H}_{t,l} \mathbf{q}_{t,l}\|_F^2 + \lambda_{t,l} \|\mathbf{s}_{t,l}\|_1 + \frac{\rho}{2} \|\mathbf{s}_{t,l} - \hat{\mathbf{s}}_{t,l}\|_F^2 + \frac{\beta_{t,l}}{2} \|\mathbf{s}_{t,l} - \mathbf{q}_{t,l} + \frac{\mathbf{p}_{t,l}}{\beta_{t,l}}\|_F^2 + \mathbf{c}_{t,l}, \quad (13)$$

Algorithm 4 Generate matrix $\mathbf{W}_k$ in (17).

Input: $k \in \mathbb{K}_N$, $\mathcal{G}_{k'} \in \mathbb{R}^{R_{1,k'} \times \dots \times R_{k'-1,k'} \times I_{k'} \times R_{k',k'+1} \times \dots \times R_{k',N}}$
for all $k' \in \mathbb{K}_N$ and $k' \neq k$
Initialization: $\mathbf{b} = (k+1 : N, 1 : k)$.
for $k' = 1$ to N and $k' \neq k$ **do**
 $\mathcal{G}_{k'} = \text{transpose}(\mathcal{G}_{k'}, \mathbf{b})$.
end for
 $c_1 = 1, d_1 = 2$, and $\mathcal{W}_k = \mathcal{G}_{\mathbf{b}(1)}$.
for $i = 1$ to $N - 2$ **do**
 $\mathcal{W}_k = \mathcal{W}_k \times_{d_1, d_2, \dots, d_i}^{c_1, c_2, \dots, c_i} \mathcal{G}_{\mathbf{b}(i+1)}$.
for $j = 1$ to $i + 1$ **do**
 $c_j = j$.
 $d_j = 2 + (j - 1)(N - i)$.
end for
end for
 $\mathcal{W}_k = \text{transpose}(\mathcal{W}_k, (2(N - k) + 1 : 2(N - 1), 1 : 2(N - k)))$.
 $\mathbf{e} = \text{ones}(1, N - 1)$ and $\mathbf{f} = \text{ones}(1, N - 1)$.
for $i = 1$ to $N - 1$ **do**
 $\mathbf{e}(i) = 2i$ and $\mathbf{f}(i) = 2i - 1$.
end for
 $\mathbf{W}_k = \text{GenUnfold}(\mathcal{W}_k, (\mathbf{e}|\mathbf{f}))$.
Output: $\mathbf{W}_k \in \mathbb{R}^{\prod_{i=1}^{k-1} R_{i,k} \times \prod_{i=k+1}^N R_{k,i} \times \prod_{i=1, i \neq k}^N I_i}$.

where $\mathbf{p}_{t,l}$ is the Lagrangian multiplier, $\beta_{t,l} > 0$ is the penalty parameter, and $\mathbf{c}_{t,l}$ is a variable that is independent of both $\mathbf{s}_{t,l}$ and $\mathbf{q}_{t,l}$. Then, we iteratively update the variables $\mathbf{q}_{t,l}$, $\mathbf{s}_{t,l}$, and $\mathbf{p}_{t,l}$ as

$$\begin{cases} \mathbf{q}_{t,l} = \underset{\mathbf{q}_{t,l}}{\text{argmin}} L_{\beta_{t,l}}(\mathbf{s}_{t,l}, \mathbf{q}_{t,l}, \mathbf{p}_{t,l}), \\ \mathbf{s}_{t,l} = \underset{\mathbf{s}_{t,l}}{\text{argmin}} L_{\beta_{t,l}}(\mathbf{s}_{t,l}, \mathbf{q}_{t,l}, \mathbf{p}_{t,l}), \\ \mathbf{p}_{t,l} = \mathbf{p}_{t,l} + \beta_{t,l}(\mathbf{s}_{t,l} - \mathbf{q}_{t,l}). \end{cases} \quad (14)$$

That is,

$$\begin{cases} \mathbf{q}_{t,l} = [\mathbf{H}_{t,l}^T \mathbf{H}_{t,l} + \beta_{t,l} \mathbf{I}]^{-1} [\mathbf{H}_{t,l}^T \mathbf{x} + \beta_{t,l} \mathbf{s}_{t,l} + \mathbf{p}_{t,l}], \\ \mathbf{s}_{t,l} = \text{shrink}\left(\frac{\rho \hat{\mathbf{s}}_{t,l} + \beta_{t,l} \mathbf{q}_{t,l} - \mathbf{p}_{t,l}}{\rho + \beta_{t,l}}, \frac{\lambda_{t,l}}{\rho + \beta_{t,l}}\right), \\ \mathbf{p}_{t,l} = \mathbf{p}_{t,l} + \beta_{t,l}(\mathbf{s}_{t,l} - \mathbf{q}_{t,l}), \end{cases} \quad (15)$$

where $[\text{shrink}(\mathbf{a}, b)](i) = \text{sign}(\mathbf{a}(i)) \max(|\mathbf{a}(i)| - b, 0)$, and $\text{sign}(\mathbf{a}(i))$ preserves the sign of $\mathbf{a}(i)$.

According to Theorem 1, once the number of non-zero entries in the diagonal matrices stabilizes, we can transform the optimization problem (7) into the following problem:

$$\min_{\bar{\mathcal{G}}} \frac{1}{2} \|\mathcal{X} - \text{FCTN}(\{\bar{\mathcal{G}}_k\}_{k=1}^N)\|_F^2 + \frac{\mu}{2} \sum_{k \in \mathbb{K}_N} \|\bar{\mathcal{G}}_k\|_F^2, \quad (16)$$

where $\bar{\mathcal{G}}_k$ for $k \in \mathbb{K}_N$ are obtained from (5). We adopt FCTN decomposition for consistency, since all TN decompositions discussed in this article can be viewed as special instances of FCTN decomposition, as demonstrated in Remark 1. Within the PAM framework, the solution to (16) can be obtained by alternately updating $\bar{\mathcal{G}}_k$ as follows:

$$\bar{\mathbf{G}}_{k(k)} = (\mathbf{X}_{(k)} \mathbf{W}_k^T + \rho \hat{\mathbf{G}}_{k(k)}) (\mathbf{W}_k \mathbf{W}_k^T + (\mu + \rho) \mathbf{I})^{-1}, \quad (17)$$

where $\mathbf{W}_k$ is computed using Algorithm 4.

In our algorithm, if the number of non-zero entries in the diagonal matrices remains unchanged for 10 consecutive iterations, we simplify problem (7) into (16). This transformation affects both optimization efficiency and solution quality. From the perspective of optimization efficiency, the diagonal

matrices in problem (7) require iterative updates through an inner ADMM procedure due to the non-smooth sparsity regularization. After the transformation, these updates are avoided, reducing the subsequent computational cost. From the perspective of solution quality, the inner ADMM procedure only provides approximate solutions within limited iterations, and the resulting errors may affect later updates. Once the TN structure has stabilized, absorbing the stable non-zero components into the TN cores allows problem (16) to directly refine the cores under the identified structure, leading to a more stable and accurate solution.

3.3 Initialization Scheme

SVDinsTN begins with a “fully-connected” topology as its initial setup, which results in substantial computational demands during early iterations when the TN ranks $R_{t,l}$ for all $(t, l) \in \mathbb{T}\mathbb{L}_N$ are initialized with large values. To alleviate this, we propose an innovative initialization strategy that efficiently reduces the initial TN rank values.

In Theorem 4, we demonstrate that for an N th-order tensor $\mathcal{X}$, an SVDinsTN exists with the TN rank $R_{t,l} \leq \min(\text{rank}(\mathbf{X}_{(t)}), \text{rank}(\mathbf{X}_{(l)}))$ for all $(t, l) \in \mathbb{T}\mathbb{L}_N$. Therefore, the initial $R_{t,l}$ can be set to $\min(\text{rank}(\mathbf{X}_{(t)}), \text{rank}(\mathbf{X}_{(l)}))$. For real-world data, this value is usually reflected by the rank of mode- (t, l) slices¹ of $\mathcal{X}$. Therefore, we initialize $R_{t,l}$ and $\mathbf{S}_{t,l}$ through a truncated SVD of mode- (t, l) slices of $\mathcal{X}$, following these two steps.

Step 1: First, calculate the mean of all mode- (t, l) slices of $\mathcal{X}$, denoted by $\mathbf{X}_{t,l}$. Then, perform SVD on $\mathbf{X}_{t,l}$ to obtain $\mathbf{s}_{t,l} \in \mathbb{R}^{\min(I_t, I_l)}$, where the entries of $\mathbf{s}_{t,l}$ represent the singular values of $\mathbf{X}_{t,l}$.

Step 2: Next, apply the shrinkage operation to $\mathbf{s}_{t,l}$ as

$$\mathbf{s}_{t,l} = \text{shrink}(\mathbf{s}_{t,l}, \gamma_0 \max(\mathbf{s}_{t,l})),$$

and remove all zero entries. Then, set $\mathbf{S}_{t,l} = \text{diag}(\mathbf{s}_{t,l})$ and define the number of entries in $\mathbf{s}_{t,l}$ as $R_{t,l}$.

In real-world applications, the shrinkage operation in Step 2 reduces the initial value of $R_{t,l}$ by setting small singular values in $\mathbf{s}_{t,l}$ to zero, thereby alleviating the computational cost during the early iterations of SVDinsTN. The parameter γ_0 only controls the initial TN rank. Since the TN structure is further refined during optimization, the method remains stable within a reasonable range of γ_0 .

In Algorithm 5, we outline the pseudocode for optimizing model (7). Below, we provide a brief analysis of the computational complexity.

Computational complexity. For simplicity, we assume that all modes of the N th-order tensor $\mathcal{X}$ have the same dimension I , and all TN ranks are equal to R with $R \leq I$. Under this setting, the computational complexities of TT, TR, and FCTN are given by

$$\mathcal{O}\left(N \sum_{k=2}^{N-2} I^k R^3 + (N-2)(I^N R^2 + I^{N-1} R^4 + R^6)\right), \quad (18)$$

$$\mathcal{O}\left(N \sum_{k=2}^{N-1} I^k R^3 + N(I^N R^2 + I^{N-1} R^4 + R^6)\right), \quad (19)$$

1. Mode- (t, l) slices are obtained by fixing all but the mode- t and the mode- l indices of a tensor [61].

Algorithm 5 PAM-based algorithm to optimize model (7).

Input: A tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ and a parameter γ .
Initialization: Initialize $\mathbf{S}_{t,l}$ and $R_{t,l}$ through the initialization scheme in Section 3.3 and let $\beta_{t,l} = 1$ for all $(t, l) \in \mathbb{TL}_N$; let $\mathcal{G}_k = 1/\sqrt{I_k}$ ones($R_{1,k}, R_{2,k}, \dots, R_{k-1,k}, I_k, R_{k,k+1}, \dots, R_{k,N}$) for all $k \in \mathbb{K}_N$, $\mu = 1$, $\rho = 0.001$, and $\Delta = 1$. Let $\tilde{\mathcal{X}} = \mathcal{X}$.
for $i = 1$ to 500 **do**
 Let $\hat{\mathcal{X}} = \tilde{\mathcal{X}}$.
 if $\Delta \neq 0$ **then** # Solve (7).
 for $k = 1$ to N **do**
 Update $\mathbf{G}_{k(k)}$ using (10).
 Let $\mathcal{G}_k = \text{Fold}(\mathbf{G}_{k(k)}, k)$.
 end for
 for $t = 1$ to $N - 1$ **do**
 for $l = t + 1$ to N **do**
 Let $\lambda_{t,l} = \gamma \max(\text{diag}(\mathbf{S}_{t,l}))(\rho + \beta_{t,l})$.
 for $j = 1$ to 5 **do**
 Update $\mathbf{q}_{t,l}$, $\mathbf{s}_{t,l}$, and $\mathbf{p}_{t,l}$ using (15).
 end for
 Delete zero entries in $\mathbf{s}_{t,l}$.
 Delete the corresponding parts of $\mathcal{G}_t$ and $\mathcal{G}_l$.
 Let $\mathbf{S}_{t,l} = \text{diag}(\mathbf{s}_{t,l})$.
 Let the size of $\mathbf{s}_{t,l}$ be $R_{t,l}^{(i)}$.
 end for
 end for
 Let $\hat{\mathcal{X}} = \text{STN}(\mathcal{G}, \mathbf{S})$.
 # TN Structure Stability Judgment.
 if $i > 10$ **then**
 Let $\Delta = \max\{|R_{t,l}^{(i)} - R_{t,l}^{(i-10)}|\}_{(t,l) \in \mathbb{TL}_N}$.
 if $\Delta = 0$ **then**
 Generate $\bar{\mathcal{G}}_k$ for all $k \in \mathbb{K}_N$ using (5).
 end if
 end if
 else # Solve (16).
 for $k = 1$ to N **do**
 Update $\bar{\mathbf{G}}_{k(k)}$ using (17).
 Let $\bar{\mathcal{G}}_k = \text{Fold}(\bar{\mathbf{G}}_{k(k)}, k)$.
 end for
 Let $\hat{\mathcal{X}} = \text{FCTN}(\{\bar{\mathcal{G}}_k\}_{k=1}^N)$.
 end if
 # Convergence Judgment.
 if $\|\hat{\mathcal{X}} - \tilde{\mathcal{X}}\|_F / \|\tilde{\mathcal{X}}\|_F < 10^{-5}$ **then**
 Break the loop.
 end if
end for
Output: $\bar{\mathcal{G}}_k$ for all $k \in \mathbb{K}_N$ (generated using (5) if $\Delta \neq 0$) and $R_{t,l}^{(i)}$ for all $(t, l) \in \mathbb{TL}_N$.

and

$$\mathcal{O}\left(N \sum_{k=2}^N I^k R^{k(N-k)+k-1} + N I^{N-1} R^{2(N-1)}\right), \quad (20)$$

respectively. SVDinsTN is built upon the FCTN framework and additionally updates the diagonal matrices $\mathbf{S}$ to reveal the underlying TN structure. Therefore, when $\mathbf{S}$ is updated separately, its computational complexity is

$$\mathcal{O}\left(N^2 \sum_{k=2}^N I^k R^{k(N-k)+k} + N^4 I R^N + N^2 I^N R^2\right), \quad (21)$$

which is higher than that of FCTN in the early iterations. Once $\mathbf{S}$ is absorbed into the TN cores $\mathcal{G}$, the computational complexity decreases to (20). In practice, the actual run time also depends on the learned TN ranks, and a compact TN structure with smaller ranks can reduce the practical computational cost.

TABLE 2

CR ($\downarrow$) and run time ($\times 1000$ s, $\downarrow$) of different methods under an RE bound of 0.01. Results are averaged over five independent tests, and “—” denotes “out of memory”.

Method	6th-order		8th-order		10th-order	
	CR	Time	CR	Time	CR	Time
TR	1.35%	0.006	0.064%	0.034	—	—
FCTN	2.13%	0.002	—	—	—	—
TNGreedy	0.88%	0.167	0.016%	2.625	0.0008%	45.39
TNGA	0.94%	3.825	0.024%	51.40	—	—
TNLS	1.11%	0.673	0.038%	59.83	—	—
TNALE	1.65%	0.201	0.047%	19.96	—	—
SVDinsTN	1.13%	0.002	0.016%	0.017	0.0007%	0.608

3.4 Numerical Results

To evaluate the performance of SVDinsTN, we conduct experiments on both synthetic and real-world datasets. The compared methods include two fixed-topology TN decomposition methods, i.e., TR [37] and FCTN [38], and four TN structure search (TN-SS) methods, i.e., TNGreedy [52], TNGA [53], TNLS [45], and TNALE [54]. The goal of this experiment is to show that SVDinsTN can significantly reduce computational time while achieving representation ability comparable to state-of-the-art TN-SS methods. Meanwhile, we also examine whether it provides stronger representation ability than fixed-topology TN decomposition methods.

To quantify the performance, each test tensor is represented by different methods, and the corresponding *compression ratio* (CR) is reported under a prescribed *representation error* (RE) bound. Here, CR is defined as $F_{\mathcal{G}}/F_{\mathcal{X}} \times 100\%$, where $F_{\mathcal{G}}$ represents the number of entries in the TN cores used to approximate the original tensor, and $F_{\mathcal{X}}$ is the total number of entries in the original tensor; RE is defined as $\|\mathcal{X} - \tilde{\mathcal{X}}\|_F / \|\mathcal{X}\|_F$, where $\mathcal{X}$ denotes the original tensor and $\tilde{\mathcal{X}}$ is the approximated tensor. Through this setup, a smaller CR indicates a more compact representation, provided that the RE satisfies the prescribed bound. In all tests, the parameter γ is selected from $[10^{-7}, 10^{-3}]$. The parameters of all compared methods are manually tuned according to their released codes to achieve their optimal performance.

Synthetic data. We generate five tensors for each of the 6th-, 8th-, and 10th-order cases. For each tensor, the TN cores $\mathcal{G}_k$ for all $k \in \mathbb{K}_N$ and the diagonal matrices $\mathbf{S}_{t,l}$ for all $(t, l) \in \mathbb{TL}_N$ are first generated, where each entry is independently sampled from the uniform distribution on $[0, 1]$. The synthetic tensor is then constructed as $\mathcal{X} = \text{STN}(\mathcal{G}, \mathbf{S})$. For each synthetic tensor, the size of each tensor mode, the number of edges in the underlying TN topology, and the rank on each edge are randomly selected from $\{5, 6, 7, 8\}$, $\{6, 8, 10\}$, and $\{2, 3\}$, respectively.

We compare SVDinsTN with the baseline methods in terms of CR and *run time*, where all methods are required to satisfy an RE bound of 0.01. The results are summarized in Tab. 2. It can be observed that SVDinsTN significantly outperforms the fixed-topology TN decomposition methods, such as TR and FCTN, in terms of CR, demonstrating that it can produce a more compact structure than predefined topologies. Furthermore, compared with TNGreedy, TNGA, TNLS, and TNALE, SVDinsTN is approximately 100 to 1000

TABLE 3

CR ($\downarrow$) and run time ($\times 1000$ s, $\downarrow$) of different methods on light field data under different RE bounds.

Method	RE: 0.01		RE: 0.05		RE: 0.1	
	CR	Time	CR	Time	CR	Time
<i>Bunny</i>						
TR	60.5%	13.54	17.4%	0.471	5.31%	0.118
FCTN	65.1%	13.08	20.9%	0.473	3.93%	0.041
TNGreedy	26.1%	11.02	6.32%	1.021	2.34%	0.362
TNGA	27.9%	1014	5.01%	180.3	2.25%	12.52
TNLS	24.3%	1402	4.26%	63.70	2.16%	24.53
TNALE	26.3%	144.5	4.52%	18.36	2.26%	3.064
SVDinsTN	<u>22.4%</u>	0.745	6.92%	0.029	2.66%	0.005
<i>Knights</i>						
TR	74.7%	10.31	26.9%	3.835	9.15%	0.423
FCTN	73.5%	12.35	20.9%	0.619	3.93%	0.014
TNGreedy	32.1%	12.53	7.55%	1.366	3.50%	0.481
TNGA	38.7%	912.9	5.01%	140.2	2.44%	12.52
TNLS	27.3%	1286	4.73%	75.51	2.15%	5.320
TNALE	27.6%	266.4	<u>4.52%</u>	25.05	<u>2.10%</u>	3.386
SVDinsTN	32.0%	1.548	5.64%	0.104	2.76%	0.019
<i>Truck</i>						
TR	62.8%	17.62	22.6%	1.738	6.00%	0.090
FCTN	69.3%	7.735	20.9%	2.953	3.93%	0.159
TNGreedy	26.9%	6.676	7.26%	1.259	3.35%	0.488
TNGA	27.9%	1029	5.01%	170.3	2.85%	14.83
TNLS	26.4%	992.6	4.99%	119.8	2.57%	19.35
TNALE	24.7%	239.3	5.77%	19.54	2.90%	5.160
SVDinsTN	<u>23.5%</u>	1.051	6.42%	0.152	2.83%	0.023

times faster while maintaining comparable CR values. This is because TNGreedy, TNGA, TNLS, and TNALE rely on a sampling-evaluation framework, which requires numerous repeated structure evaluations. In contrast, SVDinsTN does not require repeated structure sampling and evaluation. Moreover, SVDinsTN is capable of handling tensors of higher orders, up to 10th-order tensors. In contrast, all other compared methods, except TNGreedy, are unable to process tensors of this order. The underlying reason is that, in SVDinsTN, the shrinkage operation used during initialization constrains the initial TN ranks to relatively small values, thereby enhancing both storage and computational efficiency. During iterative optimization, the sparsity-inducing term further promotes a compact TN structure, leading to a more efficient representation.

Light field data. We utilize three light field datasets²: *Bunny*, *Knights*, and *Truck*, which can be represented as fifth-order tensors with dimensions $40 \times 60 \times 3 \times 9 \times 9$ (spatial height $\times$ spatial width $\times$ color channel $\times$ vertical grid $\times$ horizontal grid). Tab. 3 presents the CR and run time results of different methods on these light field datasets. It can be observed that SVDinsTN achieves lower CR values and requires shorter run time than fixed-topology TR and FCTN in most cases. Although SVDinsTN starts from a “fully-connected” topology, the shrinkage operation in its initialization scheme constrains the initial TN ranks to relatively small values, thereby reducing the computational cost in the early iterations. During iterative optimization, the sparsity-

2. <http://lightfield.stanford.edu/lfs.html>

TABLE 4

OA ($\uparrow$), AA ($\uparrow$), κ ($\uparrow$), and FLOPs ($\downarrow$) of all tensor decomposition methods on the *University of Pavia* dataset, with an approximately consistent number of parameters (Params, $\downarrow$).

Method	CP	Tucker	TT	TR	FCTN	SVDinsTN
OA	92.26	92.81	92.78	92.82	93.08	93.66
AA	94.64	94.94	94.88	94.96	94.99	95.58
κ	89.69	90.40	90.37	90.42	90.76	91.52
FLOPs	8.83G	8.60G	8.61G	17.27G	22.29G	7.84G
Params	57.40K	58.23K	57.49K	66.01K	57.40K	56.46K

inducing term further promotes a compact TN structure, leading to a more efficient representation. Compared with TN-SS methods, SVDinsTN avoids the expensive sampling-evaluation procedure for TN structure search. As a result, it achieves about 100 to 1000 times speedup over existing TN-SS methods while maintaining comparable CR values.

4 APPLICATIONS

In this section, we apply the proposed SVDinsTN to real-world applications, including neural network compression, multi-temporal remote sensing image cloud removal, and tensor robust principal component analysis. *Our objective is to demonstrate that SVDinsTN can be adapted to a wide range of tasks as easily as fixed-topology TN decomposition methods, while delivering enhanced performance.*

4.1 Neural Network Compression

Neural network compression leverages tensor decomposition to approximate the low-rank structure of convolutional layer kernels, thereby reducing model complexity. We apply various tensor decomposition methods, including CP decomposition [10], Tucker decomposition [10], TT decomposition [36], TR decomposition [37], FCTN decomposition [38], and the proposed SVDinsTN, to compress convolutional layers with the same 5×5 kernel size in the hyperspectral image (HSI) classification network DilatedFCN [62]. We employ two HSI datasets, i.e., the *University of Pavia* dataset³ and the *Houston* dataset⁴, with 300 randomly selected samples for training and the rest for testing. *University of Pavia* contains 9 land-cover classes, covers 103 spectral bands, and has a spatial size of 610×340 pixels. *Houston* contains 15 land-cover classes, covers 144 spectral bands, and has a spatial size of 349×1905 pixels. In this test, the parameter γ is set to 0.001 and 0.002 on the first and second datasets, respectively. The parameters of all compared methods are manually adjusted to achieve their best performance.

For implementation, Adam is used for optimization with an initial learning rate of 0.005. Each model is trained for 2000 epochs, with other settings aligned with [62]. We employ the *overall accuracy* (OA), *average accuracy* (AA), and *Kappa coefficient* (κ) to quantify classification performance, and apply *floating-point operations* (FLOPs) and the *number*

3. https://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes

4. <http://dase.grss-ieee.org/>

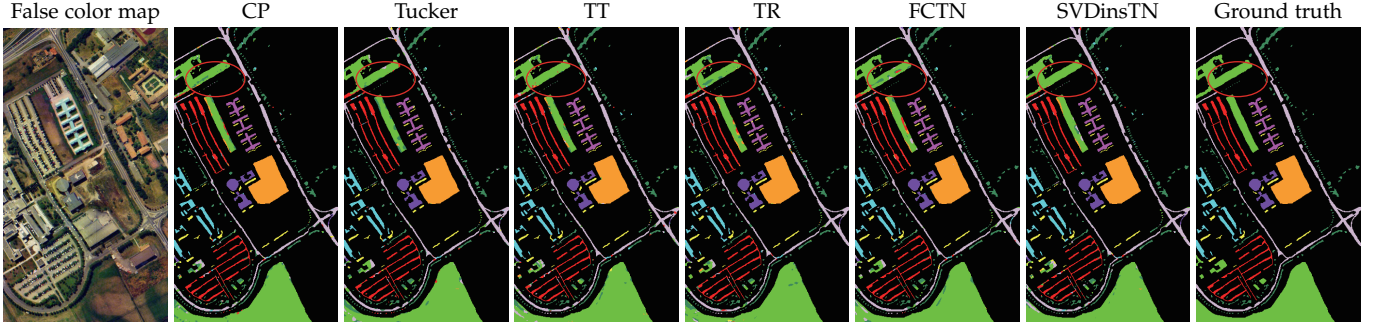


Fig. 3. Classification maps generated using various tensor decomposition methods on the *University of Pavia* dataset.

TABLE 5

OA ($\uparrow$), AA ($\uparrow$), κ ($\uparrow$), and FLOPs ($\downarrow$) of all tensor decomposition methods on the *Houston* dataset, with an approximately consistent number of parameters (Params, $\downarrow$).

Method	CP	Tucker	TT	TR	FCTN	SVDinsTN
OA	95.01	95.59	95.22	95.62	95.93	96.05
AA	96.13	96.64	96.30	96.52	96.85	96.96
κ	94.56	95.18	94.79	95.23	95.56	95.69
FLOPs	16.43G	16.13G	16.14G	22.89G	30.28G	12.43G
Params	32.78K	33.82K	33.51K	32.78K	32.78K	32.99K

of parameters (Params) to measure the model complexity. All experiments are repeated 20 times to reduce potential bias.

Result analysis. Tabs. 4 and 5 present the OA, AA, κ , Params, and FLOPs for different methods on the *University of Pavia* and *Houston* datasets, respectively. As shown, SVDinsTN achieves *higher accuracy with fewer FLOPs*, while maintaining a similar number of parameters compared to other methods. Notably, compared to TR and FCTN decompositions, SVDinsTN demonstrates a particularly significant computational advantage on the *University of Pavia* dataset. For a more intuitive comparison, we show in Fig. 3 the classification maps generated by various tensor decomposition methods on the *University of Pavia* dataset. We observe that the classification map produced by SVDinsTN closely resembles the ground truth, especially in the green-colored class regions. This success can be attributed to the structure-learning capability of SVDinsTN, which allows it to better capture the underlying structure of the data.

4.2 Remote Sensing Image Cloud Removal

Multi-temporal remote sensing image (MTRSI) cloud removal aims to eliminate cloud cover from images, which is essential for enhancing observation accuracy and data completeness, enabling more reliable surface analysis. Assuming the cloud locations are known, the SVDinsTN-based cloud removal model can be formulated as follows:

$$\begin{aligned}
 \min_{\mathcal{X}, \mathcal{G}, \mathbf{S}} \quad & \frac{1}{2} \|\mathcal{X} - \text{STN}(\mathcal{G}, \mathbf{S})\|_F^2 + \frac{\mu}{2} \sum_{k \in \mathbb{K}_N} \|\mathcal{G}_k\|_F^2 \\
 & + \sum_{(t,l) \in \text{TL}_N} \lambda_{t,l} \|\mathbf{S}_{t,l}\|_1, \\
 \text{s.t.} \quad & \mathcal{P}_\Omega(\mathcal{X} - \mathcal{F}) = 0,
 \end{aligned} \tag{22}$$

where $\mathcal{F} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ is the observed MTRSI, $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ is the underlying clean MTRSI, Ω is the in-

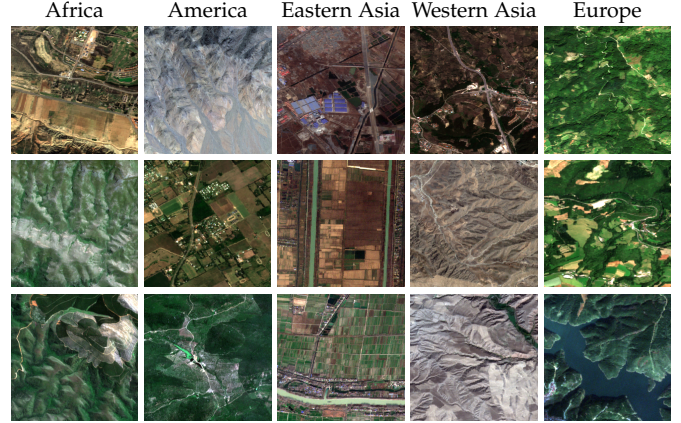


Fig. 4. The MTRSI (shown in RGB) used for experiments.

dex set of the entries that are not covered by clouds, and $\mathcal{P}_\Omega(\mathcal{X})$ is a projection operator that keeps the entries in Ω unchanged and projects the remaining entries to zeros.

We also use PAM to solve (22). The solutions to the $\mathcal{G}_k$ -subproblem and $\mathbf{S}_{t,l}$ -subproblem are the same as those given in Section 3.2. The $\mathcal{X}$ -subproblem can be solved by

$$\mathcal{X} = \mathcal{P}_{\Omega^c}((\text{STN}(\mathcal{G}, \mathbf{S}) + \rho \hat{\mathcal{X}})/(1 + \rho)) + \mathcal{P}_\Omega(\mathcal{F}), \tag{23}$$

where $\hat{\mathcal{X}}$ is the result at the previous iteration and the initial $\mathcal{X}$ is the observed tensor $\mathcal{F}$.

We employ 15 MTRSI from different regions, i.e., *Africa*, *America*, *Eastern Asia*, *Western Asia*, and *Europe*, and each region includes three MTRSI [63] shown in Fig. 4. All test MTRSI have a spatial size of 256×256 pixels, cover 13 spectral bands, and contain 6 temporal nodes. We compare five representative tensor decomposition-based methods, i.e., BCPF [2], TMac [64], TMacTT [39], TRLRF [49], and TW [65]. We assume that the simulated clouds occupy the same position across all bands at the same temporal node. We select the first three temporal nodes to introduce clouds, with the cloud masks (locations) extracted from real cloudy images. We employ the *peak signal-to-noise ratio (PSNR)* to evaluate the reconstruction performance. For each test MTRSI, the PSNR values are averaged over all spectral bands and the temporal nodes affected by cloud contamination. In all tests, the parameter γ is set to 0.0005. The parameters of all compared methods are manually adjusted to achieve their best performance.

Result analysis. Tab. 6 reports the PSNR and run time results of different methods on MTRSI. It can be observed

TABLE 6
PSNR (mean±standard deviation, †) and run time (mean±standard deviation, ‡) of different methods on MTRSI, where the mean and standard deviation are calculated from the results of three test MTRSI in each region.

Region	Index	Observed	BCPF	TMac	TMacTT	TRLRF	TW	SVDinsTN
Africa	PSNR	13.395±1.229	34.480±1.581	34.496±4.517	35.760±4.498	36.489±1.912	35.095±7.128	39.231±3.331
	Time (s)	—	704.90±158.3	61.775±25.97	30.746±14.68	250.03±125.4	839.68±199.4	820.91±31.44
America	PSNR	13.756±1.689	36.989±2.641	36.991±1.788	39.177±2.858	38.736±2.370	39.982±3.714	42.045±3.317
	Time (s)	—	721.47±84.54	74.643±32.70	41.531±10.40	187.292±64.48	712.98±208.1	681.72±261.5
Eastern Asia	PSNR	13.362±1.017	33.683±1.273	32.939±3.094	34.332±2.609	35.277±0.922	33.568±0.347	36.609±1.696
	Time (s)	—	608.01±89.32	34.215±8.593	17.503±1.529	226.60±101.2	763.17±173.5	783.74±356.5
Western Asia	PSNR	13.583±1.826	34.411±0.784	34.533±1.098	36.187±1.055	36.535±1.357	36.759±2.035	38.506±2.533
	Time (s)	—	687.84±67.41	51.251±26.14	25.282±13.11	228.39±133.1	758.49±241.8	646.09±162.2
Europe	PSNR	12.308±0.288	33.180±0.813	33.194±2.419	34.449±2.101	35.084±1.011	34.785±1.561	36.260±0.736
	Time (s)	—	761.74±100.4	61.769±25.60	40.963±37.32	406.66±420.4	861.03±51.76	851.20±93.08

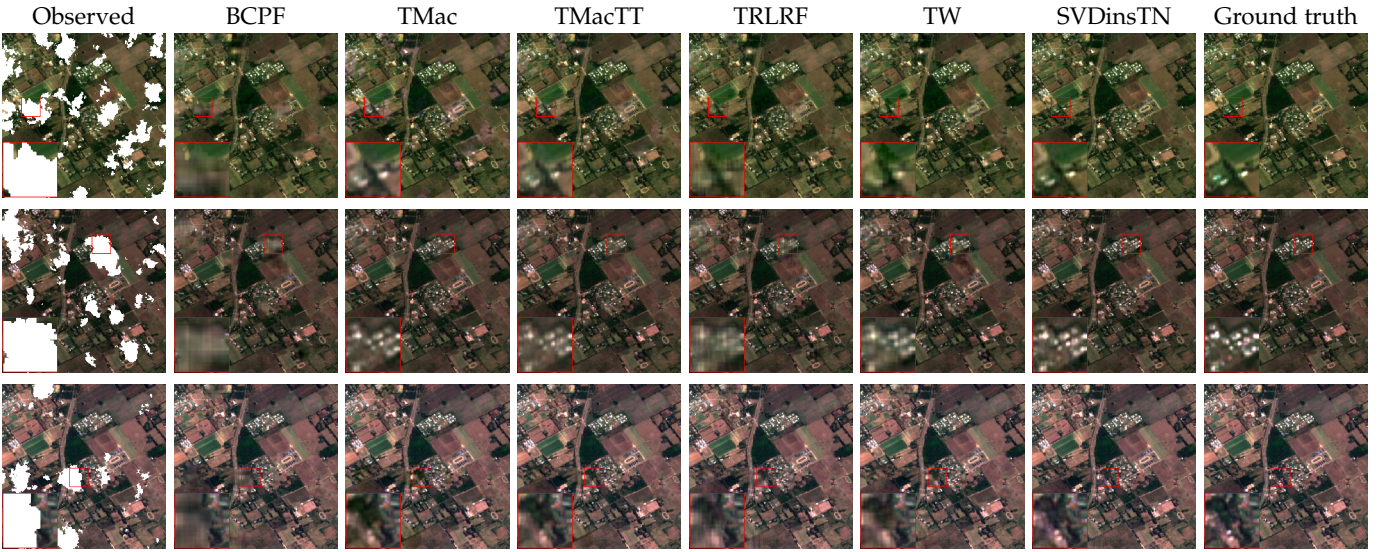


Fig. 5. Reconstructed MTRSI (shown in RGB) obtained by different methods in *America* at the temporal nodes 1, 2, and 3, respectively.

that SVDinsTN consistently obtains the best *PSNR* values in all regions, indicating its strong restoration ability for different MTRSI. Specifically, SVDinsTN improves the *PSNR* over the second-best method by about 2 dB. For run time, SVDinsTN is not always the fastest method, since it needs to learn a compact TN structure during optimization. Nevertheless, its run time remains comparable to BCPF and TW in most cases. These results show that SVDinsTN achieves better reconstruction quality with an acceptable run time. For visual comparison, Fig. 5 displays the reconstructed results produced by different methods on the second MTRSI in *America*. As observed, SVDinsTN is more effective in capturing finer details and maintaining high color fidelity. This demonstrates the superiority of SVDinsTN in producing more accurate and visually consistent results compared to other methods. These results also indicate that SVDinsTN is capable of revealing a TN structure that is better adapted to the test MTRSI than existing fixed TN structures.

4.3 Tensor Robust Principal Component Analysis

Tensor robust principal component analysis (TRPCA) aims to decompose a higher-order tensor into low-rank and

sparse components, which is essential for removing sparse noise and detecting anomalies in high-dimensional data. Mathematically, the SVDinsTN-based TRPCA model can be formulated as follows:

$$\begin{aligned}
 \min_{\mathcal{X}, \mathcal{G}, \mathbf{S}, \mathcal{E}} \quad & \frac{1}{2} \|\mathcal{X} - \text{STN}(\mathcal{G}, \mathbf{S})\|_F^2 + \frac{\mu}{2} \sum_{k \in \mathbb{K}_N} \|\mathcal{G}_k\|_F^2 \\
 & + \sum_{(t,l) \in \mathbb{TL}_N} \lambda_{t,l} \|\mathbf{S}_{t,l}\|_1 + \alpha \|\mathcal{E}\|_1, \\
 \text{s.t. } \quad & \mathcal{Y} = \mathcal{X} + \mathcal{E},
 \end{aligned} \tag{24}$$

where $\mathcal{Y} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ is the observed high-dimensional data, $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ is the low-rank component, $\mathcal{E} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ is the sparse component, and α is a balance parameter. By introducing the penalty term $\frac{\eta}{2} \|\mathcal{Y} - (\mathcal{X} + \mathcal{E})\|_F^2$, (24) can be relaxed into the following penalized form:

$$\begin{aligned}
 \min_{\mathcal{X}, \mathcal{G}, \mathbf{S}, \mathcal{E}} \quad & \frac{1}{2} \|\mathcal{X} - \text{STN}(\mathcal{G}, \mathbf{S})\|_F^2 + \frac{\eta}{2} \|\mathcal{Y} - (\mathcal{X} + \mathcal{E})\|_F^2 \\
 & + \frac{\mu}{2} \sum_{k \in \mathbb{K}_N} \|\mathcal{G}_k\|_F^2 + \sum_{(t,l) \in \mathbb{TL}_N} \lambda_{t,l} \|\mathbf{S}_{t,l}\|_1 + \alpha \|\mathcal{E}\|_1.
 \end{aligned} \tag{25}$$

We also use PAM to solve (25). The solutions to the $\mathcal{G}_k$ -subproblem and $\mathbf{S}_{t,l}$ -subproblem are the same as those

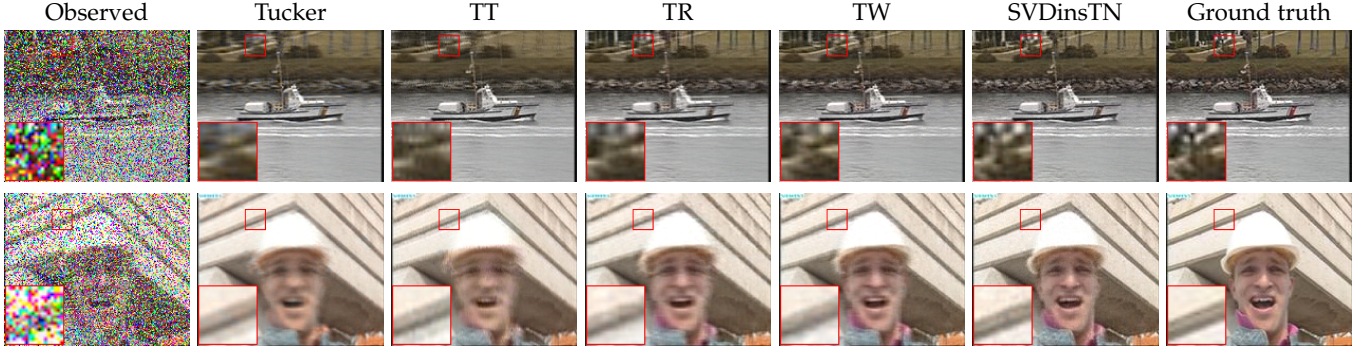


Fig. 6. Recovered results obtained by different methods for the 25th frame of the *Coastguard* and *Foreman* videos, under a noise level of 0.4.

TABLE 7
PSNR ($\uparrow$) and run time ($\downarrow$) of different methods on color videos.

Method	Noise level: 0.3		Noise level: 0.4		Noise level: 0.5	
	PSNR	Time (s)	PSNR	Time (s)	PSNR	Time (s)
<i>News</i>						
Observed	10.011	—	8.7287	—	7.7556	—
Tucker	32.728	192.78	30.152	114.79	29.427	101.29
TT	31.546	168.90	29.439	87.926	29.023	88.190
TR	34.436	2293.4	31.446	697.01	30.646	707.78
TW	34.583	1201.2	32.974	776.46	31.503	297.60
SVDinsTN	36.296	776.46	34.757	551.94	32.538	345.06
<i>Coastguard</i>						
Observed	10.552	—	9.2652	—	8.2985	—
Tucker	28.748	175.10	27.291	101.12	26.672	105.92
TT	27.782	162.04	26.610	86.968	25.868	79.661
TR	30.846	2315.0	28.541	701.64	26.993	706.49
TW	30.458	1178.2	29.325	1007.5	27.655	646.89
SVDinsTN	33.288	518.09	30.952	339.00	29.319	333.44
<i>Foreman</i>						
Observed	10.334	—	9.0701	—	8.1013	—
Tucker	30.642	189.82	28.584	185.76	28.066	105.65
TT	28.861	162.67	27.228	158.21	26.571	83.978
TR	31.521	1619.4	28.743	943.17	27.950	704.46
TW	31.763	967.11	30.232	1083.3	29.088	670.24
SVDinsTN	34.975	754.14	32.902	460.43	30.099	353.92
<i>Suzie</i>						
Observed	10.809	—	9.5252	—	8.5731	—
Tucker	31.436	196.36	30.191	102.70	29.284	100.91
TT	29.778	163.29	28.430	80.781	27.789	84.822
TR	35.360	2306.4	32.774	699.59	32.042	702.23
TW	35.036	1239.8	33.321	813.54	31.914	685.69
SVDinsTN	36.762	870.96	34.677	395.78	33.036	325.85

given in Section 3.2. The $\mathcal{E}$ -subproblem can be solved by

$$\mathcal{E} = \text{shrink}\left(\frac{\rho\hat{\mathcal{E}} + \eta(\mathcal{Y} - \mathcal{X})}{\rho + \eta}, \frac{\alpha}{\rho + \eta}\right), \quad (26)$$

where $\hat{\mathcal{E}}$ is the result at the previous iteration and the initial $\mathcal{E}$ is a zero tensor. The $\mathcal{X}$ -subproblem can be solved by

$$\mathcal{X} = (\text{STN}(\mathcal{G}, \mathbf{S}) + \eta(\mathcal{Y} - \mathcal{E}) + \rho\hat{\mathcal{X}})/(1 + \eta + \rho), \quad (27)$$

where $\hat{\mathcal{X}}$ is the result at the previous iteration and the initial $\mathcal{X}$ is the observed tensor $\mathcal{Y}$.

We use the task of color video denoising as an example to evaluate the performance of the proposed SVDinsTN-based TRPCA method. We test four color videos⁵, i.e., *News*, *Coastguard*, *Foreman*, and *Suzie*. All test color videos have a spatial size of 144×176 pixels and contain 50 frames. For each test video, we randomly add salt-and-pepper noise with noise levels of 0.3, 0.4, and 0.5. We compare four representative tensor decomposition-based methods, namely Tucker [64], TT [66], TR [67], and TW [65]. We evaluate the reconstructed results using PSNR, whose values are averaged over all frames. In all tests, the parameter γ is selected from $[0.0001, 0.001]$. The parameters of all compared methods are manually adjusted to achieve their best performance.

Result analysis. Tab. 7 presents the PSNR and run time results obtained by different methods on color videos under different noise levels. As observed, SVDinsTN achieves the highest PSNR values in all test cases, demonstrating its superior reconstruction accuracy and robustness to different noise levels. Although SVDinsTN is not always the fastest method, its run time remains comparable to that of fixed-topology TN-based methods such as TR and TW. These results indicate that the adaptive learning of both TN topology and TN ranks does not introduce prohibitive computational overhead. For visual comparison, Fig. 6 shows the recovered 25th frames of the *Coastguard* and *Foreman* videos obtained by different methods under a noise level of 0.4. As observed, SVDinsTN produces better visual results than the other methods, with clearer structural details. These results further demonstrate its ability to identify a TN structure that better matches the intrinsic characteristics of the data than existing fixed TN structures.

5 DISCUSSIONS

5.1 Influence of Different Sparsity-Inducing Terms

The sparsity-inducing term in the energy function in (6) is not limited to the ℓ_1 -norm. It provides a flexible framework for exploring other sparsity-inducing terms. In this subsection, we discuss the influence of three sparsity-inducing terms on SVDinsTN, including the ℓ_1 -norm, the nonconvex log-norm, and the ℓ_0 -norm. For a matrix $\mathbf{X} \in \mathbb{R}^{m \times n}$, the log-norm is defined as

$$\|\mathbf{X}\|_{\log} = \sum_{i=1}^m \sum_{j=1}^n \log(\epsilon + |\mathbf{X}(i, j)|), \quad (28)$$

5. <http://trace.eas.asu.edu/yuv/>

TABLE 8

CR ($\downarrow$) and run time ($\times 1000s$, $\downarrow$) for SVDinsTN with various sparsity-inducing terms on light field data at different RE bounds.

Method	RE: 0.01		RE: 0.05		RE: 0.1	
	CR	Time	CR	Time	CR	Time
<i>Bunny</i>						
SVDinsTN- ℓ_1	22.4%	0.745	6.92%	0.029	2.66%	0.005
SVDinsTN- $\ell_{\log}$	21.0%	0.712	5.91%	0.011	2.60%	0.005
SVDinsTN- ℓ_0	20.7%	0.417	5.91%	0.010	2.63%	0.005
<i>Knights</i>						
SVDinsTN- ℓ_1	32.0%	1.548	5.64%	0.104	2.76%	0.019
SVDinsTN- $\ell_{\log}$	25.6%	0.553	6.02%	0.082	2.13%	0.015
SVDinsTN- ℓ_0	24.9%	0.470	5.28%	0.083	2.33%	0.029
<i>Truck</i>						
SVDinsTN- ℓ_1	23.5%	1.051	6.42%	0.152	2.83%	0.023
SVDinsTN- $\ell_{\log}$	25.2%	0.626	6.42%	0.105	3.20%	0.004
SVDinsTN- ℓ_0	24.1%	0.182	6.34%	0.065	3.20%	0.008

and the ℓ_0 -norm is defined as

$$\|\mathbf{X}\|_0 = \text{count}(\{(i, j) \mid \mathbf{X}(i, j) \neq 0\}), \quad (29)$$

where $\epsilon = 1$, and $\text{count}(\cdot)$ returns the number of entries.

We replace the ℓ_1 -norm $\|\mathbf{S}_{t,l}\|_1$ in problem (7) with the log-norm $\|\mathbf{S}_{t,l}\|_{\log}$ and the ℓ_0 -norm $\|\mathbf{S}_{t,l}\|_0$, respectively. Accordingly, the solution to the $\mathbf{S}_{t,l}$ -subproblem in (15) is modified as follows.

For the log-norm-based method, we have

$$\mathbf{s}_{t,l} = \text{shrink}_{\log}^{\epsilon} \left(\frac{\rho \hat{\mathbf{s}}_{t,l} + \beta_{t,l} \mathbf{q}_{t,l} - \mathbf{p}_{t,l}}{\rho + \beta_{t,l}}, \frac{\lambda_{t,l}}{\rho + \beta_{t,l}} \right), \quad (30)$$

where

$$[\text{shrink}_{\log}^{\epsilon}(\mathbf{a}, b)](i) = \begin{cases} \text{sign}(\mathbf{a}(i)) \frac{c_1 + \sqrt{c_2}}{2}, & \text{if } c_2 > 0, \\ 0, & \text{if } c_2 \leq 0, \end{cases}$$

with

$$c_1 = |\mathbf{a}(i)| - \epsilon, \quad c_2 = c_1^2 - 4(b - \epsilon|\mathbf{a}(i)|).$$

For the ℓ_0 -norm-based method, we have

$$\mathbf{s}_{t,l} = \text{shrink}_0 \left(\frac{\rho \hat{\mathbf{s}}_{t,l} + \beta_{t,l} \mathbf{q}_{t,l} - \mathbf{p}_{t,l}}{\rho + \beta_{t,l}}, \frac{\lambda_{t,l}}{\rho + \beta_{t,l}} \right), \quad (31)$$

where

$$[\text{shrink}_0(\mathbf{a}, b)](i) = \begin{cases} \mathbf{a}(i), & \text{if } |\mathbf{a}(i)| > \sqrt{2b}, \\ 0, & \text{if } |\mathbf{a}(i)| \leq \sqrt{2b}. \end{cases}$$

We evaluate the influence of different sparsity-inducing terms by taking the light field data representation task as an example. Tab. 8 reports the CR and *run time* of SVDinsTN with different sparsity-inducing terms on light field data under different RE bounds. It is observed that the ℓ_0 -norm-based method achieves lower CR and shorter *run time* in most cases. This is because the ℓ_0 -norm directly penalizes the number of non-zero components, thereby removing redundant rank-related components more effectively. As a result, it can obtain a more compact TN representation, which reduces the storage and computational cost.

TABLE 9

Influence of the proximal parameter ρ on PSNR for the *News* and *Foreman* videos at a noise level of 0.4.

	ρ	0	10^{-4}	10^{-3}	10^{-2}	10^{-1}	1
<i>News</i> PSNR	34.134	34.710	34.757	34.635	34.722	34.026	
<i>Foreman</i> PSNR	32.253	32.721	32.902	33.096	32.762	32.423	

TABLE 10

Influence of the regularization parameter μ on PSNR for the *News* and *Foreman* videos at a noise level of 0.4.

	μ	0.05	0.1	0.5	1	5	10
<i>News</i> PSNR	34.552	34.536	34.472	34.757	34.534	27.816	
<i>Foreman</i> PSNR	32.449	32.621	32.512	32.902	32.833	22.253	

5.2 Influence of Parameter Settings

In this subsection, we discuss several key parameter settings of SVDinsTN and their influence on reconstruction performance and computational efficiency, including the proximal parameter ρ , the regularization parameter μ , the initialization parameter γ_0 , the number of inner ADMM iterations, the maximum number of outer iterations, and the parameters used in the TRPCA task. To ensure consistency, we use the TRPCA task as a representative example for all parameter analyses.

The proximal parameter ρ controls the strength of the proximal term. When $\rho = 0$, the proximal term is removed, and PAM reduces to standard alternating minimization (AM). Tab. 9 reports the influence of ρ on the *News* and *Foreman* videos at a noise level of 0.4. It can be observed that $\rho = 0$ leads to lower PSNR values, which indicates that the standard AM update is less stable for the proposed model. When ρ is selected within a moderate range, the PSNR values remain relatively stable. This suggests that the proposed method is not highly sensitive to ρ once the proximal term is properly introduced. When ρ becomes too large, the update may become overly conservative, which can slightly degrade the reconstruction performance. Therefore, we set $\rho = 10^{-3}$ in the main experiments.

The regularization parameter μ controls the strength of the regularization imposed on the TN cores. Tab. 10 reports the influence of μ on the *News* and *Foreman* videos at a noise level of 0.4. It can be observed that the proposed method maintains stable performance within a reasonable range of μ (i.e., 0.05 to 5). Therefore, $\mu = 1$ is adopted in the main experiments.

The parameter γ_0 controls the shrinkage threshold in the initialization scheme, which determines the initial rank. Tab. 11 reports the influence of γ_0 on the *News* and *Foreman* videos at a noise level of 0.4. It can be observed that the proposed method remains stable when γ_0 is within a reasonable range (i.e., 10^{-7} to 10^{-3}). However, when γ_0 becomes too large (i.e., 10^{-2}), the performance degrades significantly because excessive shrinkage results in an overly small initial rank, limiting the representation capacity of the model. In the main experiments, we set $\gamma_0 = \gamma$, and γ is selected empirically for different tasks.

The number of inner ADMM iterations controls the accu-

TABLE 11
Influence of the initialization parameter γ_0 on PSNR for the *News* and *Foreman* videos at a noise level of 0.4.

	γ_0	10^{-7}	10^{-6}	10^{-5}	10^{-4}	10^{-3}	10^{-2}
<i>News</i>	PSNR	34.505	34.478	34.650	34.443	34.382	26.181
<i>Foreman</i>	PSNR	32.960	32.801	32.738	32.807	32.375	21.007

TABLE 12
Influence of the number of inner ADMM iterations on PSNR and run time for the *News* and *Foreman* videos at a noise level of 0.4.

<i>News</i>							
Iterations	1	2	3	4	5	6	7
PSNR	10.454	32.781	33.929	34.604	34.757	34.458	34.434
Time (s)	153.15	246.35	399.93	429.02	551.94	552.13	557.15
<i>Foreman</i>							
Iterations	1	2	3	4	5	6	7
PSNR	18.363	31.278	32.172	32.643	32.902	32.934	33.199
Time (s)	135.55	482.69	429.03	466.47	460.43	457.36	597.56

racy of solving the $S_{t,l}$ -subproblem in each outer iteration. To analyze its influence, we vary the number of inner ADMM iterations from 1 to 7. The results are reported in Tab. 12. When only one inner iteration is used, the subproblem is insufficiently optimized, leading to clearly degraded PSNR. When the number of inner iterations increases to 4 or more, the PSNR becomes relatively stable. Further increasing the number of inner iterations does not bring significant performance gains, but generally increases the run time. Therefore, we set the number of inner ADMM iterations to 5 in the main experiments.

We also analyze the influence of the maximum number of outer iterations. Fig. 7 shows the PSNR and relative error curves over 1000 iterations on the *News* and *Foreman* videos at a noise level of 0.4, where the relative error is computed between the reconstructed videos in consecutive iterations. The PSNR increases rapidly in the initial stage and becomes stable after approximately 500 iterations. Meanwhile, the relative error decreases quickly and becomes negligible around 500 iterations. These observations indicate that 500 outer iterations are sufficient for convergence in our experiments. Further iterations bring little performance improvement but increase computational time. Thus, the maximum number of outer iterations is set to 500 in the main experiments.

For the TRPCA task, we conduct a sensitivity analysis to evaluate the influence of the parameters α and η . Specifically, α is tested over $\{0.001, 0.0025, 0.005, 0.0075, 0.01, 0.025, 0.05, 0.075, 0.1\}$, and η is tested over $\{0.1, 0.25, 0.5, 0.75, 1, 2.5, 5, 7.5, 10\}$. The corresponding PSNR results on the *News* and *Foreman* videos are reported in Fig. 8. As observed, either too small or too large values of α and η can degrade the reconstruction quality, whereas moderate values generally yield higher PSNR values. Based on this analysis, we set $\alpha = 0.01$ and $\eta = 1$ in the main TRPCA experiments.

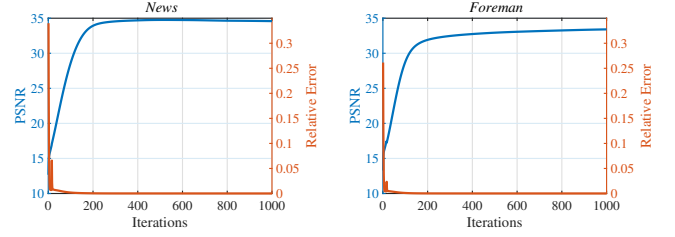


Fig. 7. PSNR and relative error curves versus the iteration number for the *News* and *Foreman* videos at a noise level of 0.4.

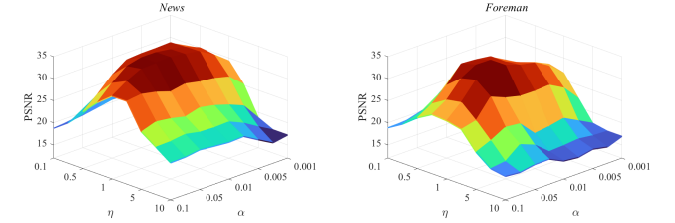


Fig. 8. Sensitivity analysis of the proposed method with respect to α and η on the *News* and *Foreman* videos at a noise level of 0.4.

6 CONCLUSION

We proposed a structure-revealing TN paradigm, termed SVDinsTN, which addresses the TN-SS problem from an energy-based perspective. Specifically, we constructed an energy function that balances the representation error and the complexity of the TN structure. This function enables the joint optimization of data representation and the discovery of a compact TN structure. As a result, SVDinsTN achieves approximately 100 to 1000 times acceleration over state-of-the-art TN-SS methods, while maintaining comparable representation capability. We also introduced SVDinsTN-based models for various low-level vision and machine learning tasks, including neural network compression, multi-temporal remote sensing image cloud removal, and tensor robust principal component analysis, demonstrating its versatility and effectiveness as a core tool for these tasks.

SVDinsTN still has two main limitations. First, the theoretical guarantee for obtaining the globally optimal TN topology and ranks remains an open problem. Second, SVDinsTN is built on a discrete tensor representation, and thus cannot be directly applied to irregularly sampled data or data defined on non-grid domains, such as point cloud data and spatial transcriptomics data. In future work, we will study theoretical guarantees for the learned TN structure and extend SVDinsTN toward continuous representations for irregular data.

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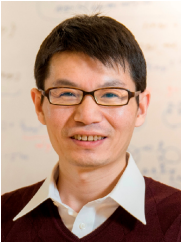


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